Nested Elimination and Half-Rate Bounds for Storage Codes

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Abstract

Can a deterministic multitape Turing machine transpose a $d \times d$ matrix of $\lceil \log_2 d \rceil$-bit entries in time linear in the input length? Earlier papers of this series reduce the impossibility of oblivious linear-time transposers to depth reduction and an all-code question about storage codes: on bounded-degree recovery relations without short paired paths, can a storage code beat rate $\frac{1}{2}$? We prove the exact half bound $2h(V) \le \sum_v h(\{v\})$ for every valid polymatroid, and hence $|C|^2 \le |\alpha|^{|V|}$ for every storage code, linear or not, over every alphabet, on large classes of recovery relations, most of them defined by the absence of transitive triangles. The engine is a nested elimination lemma (Lemma E), which extends the inequality form of an exact partition identity of the series' circular memo by outside parents with nested charges, and a decomposition theorem (Theorem D) that composes certified pieces. Its instances include triangle-free graphs of maximum degree three (Theorem N3); loopless relations without transitive triangles whose underlying graph has maximum degree three (Theorem U3), or whose vertices of in-degree at least three induce an acyclic sub-relation, with no degree bound (Theorem T); loopless relations partitioned into blocks with private outside parents, each block chordless subcubic, a chorded digon path without transitive triangles, or internally certified (Theorem PC); and every Cayley relation $\mathrm{Cay}(G, \{s, s^{-1}, t\})$ without transitive triangles, for every finite group, when $\{s, s^{-1}, t\}$ has three elements and does not contain 1 (Corollary CC). On triangle-free graphs of maximum degree three the guessing number equals the fractional matching number, and the broadcast rate of index coding is the number of vertices minus it. We also show where the method stops: in a 15-vertex relation with in- and out-degrees at most three the unique minimum nonempty core admits no certificate, some chorded cycles admit no valid sweep, the Petersen graph admits no global certificate, and on the folded cubes, whose odd girth equals their degree, the half bound fails, so that a half bound assuming only the absence of short paired paths up to a fixed length must restrict the degree. The numbered results are formalized in Lean 4, except for the step from the block form to the strongly-connected-component form of Corollary T_{int}, the digon-component form of Theorem PC, a paid-deletion corollary, a lemma and two propositions of a few lines, and the two remarks that pass from the formal code bounds to the guessing number and the broadcast rate, all of which have complete written proofs; the computations about the 15-vertex relation, the chorded cycles and the Petersen graph are numerical remarks, and the folded-cube formula is cited. These are unconditional storage-code theorems: they constrain constructions and locate the remaining work for an impossibility proof, but they do not decide whether linear-time transposition is possible.

Contributions. Hans Brende chose the problem, fixed the verification standard, and directed the research. Claude Opus 5.5, a large language model developed by Anthropic, developed the mathematics, wrote the Lean formalization, and wrote this paper under Hans Brende's direction. The engine of the paper, the nested elimination lemma, generalizes the exact partition identity of the circular memo by Hans Brende and GPT-6 Astra (OpenAI) [13, Theorem 2.1], and the treatment of graphs of circular chromatic number below three reuses that memo's Andrásfai elimination order.

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1 Introduction

1.1 The problem

Can a deterministic multitape Turing machine transpose a $d \times d$ matrix with $b(d) = \max(1, \lceil \log_2 d \rceil)$-bit entries, written row by row, in time linear in the input length $L = d^2 b(d)$? This is the statement (T) of the first paper of this series [12]. By a reduction of Harvey and van der Hoeven [24, Cor. 6.9], a linear-time multiplication machine would give a linear-time transposer, so a proof of ¬(T) would give the first superlinear lower bound for off-line integer multiplication on multitape Turing machines. The question is open, and this paper does not decide it.

For oblivious machines the series has reduced ¬(T) to two statements. The fourth paper [11, Theorem 4.10 and Corollary 4.11] shows that depth reduction of the time–space dependency graph by a cheap genie, together with an index-coding lower bound on the side information of the remaining bounded-depth cones (the *crux*), rules out oblivious linear-time transposers. *Directed Recovery Structures in Tape Computations* [15] shows that the cone relations of linear-time schedules have few coordinates whose input and demand ports are close in the paired incidence graph, and that an affirmative answer to one machine-free question would supply the crux on those cone relations [15, Question 9.1 and Proposition 9.2]: do storage codes on bounded-degree recovery relations with this paired-distance separation have rate at most $\frac{1}{2} + o(1)$, for every alphabet?

This is a storage-code paper, like the series paper *Covers and Local Generation in Linear Storage Codes* [14]. We prove the exact half bound for all codes on large classes of recovery relations without short paired paths, and show where the method stops. A recovery relation assigns to each coordinate v the set $N(v)$ of coordinates from which it must be recoverable. For a polymatroid h valid for N, that is, with $h(N(v) \cup \{v\}) = h(N(v))$ for every v, the bound reads

$$2h(V) \;\le\; \sum_{v \in V} h(\{v\}). \tag{$\star$}$$

Applied to the entropy of a uniformly random codeword, ($\star$) gives $|C|^2 \le |\alpha|^{|V|}$ for every storage code $C \subseteq \alpha^V$, linear or not, over every alphabet [11]: no code beats the pairing. Our hypotheses are exact rather than asymptotic. The central one is that the relation has no *transitive triangle* $x \to y \to z$, $x \to z$, where $u \to v$ means $u \in N(v)$; for loopless relations this says exactly that every coordinate has paired distance at least five (Lemma 2.1).

1.2 Results

The engine (Section 3). *Lemma E* (nested elimination) eliminates a piece H one vertex at a time. Each vertex must be recovered from the vertices already eliminated, from one record other than its own, and possibly from one vertex outside the piece; steps that use the same outside vertex must be nested. Then $2h(H) \le \sum_i h(P_i) + \sum_{w \notin H} I(w; H)$ (Lemma 3.2). Without outside vertices this is the inequality form of the exact partition identity of the circular memo [13, Theorem 2.1]; the nested charging is new. *Theorem D* (decomposition) contracts certified pieces one after another, each in the relation left by the earlier ones, and proves ($\star$) whenever the vertex set has such a decomposition (Theorem 3.4). Relative 2-dicolourings (Lemma B_{dir}, Lemma 3.7) and global certificates, which pull back along homomorphisms (Lemma 3.10), are the basic pieces.

The main theorem. Every relation in the following classes has a decomposition, or a global certificate, and hence satisfies ($\star$); every storage code on it satisfies $|C|^2 \le |\alpha|^{|V|}$.

(a) 2-dicolourable relations, in particular bipartite graphs (Proposition 4.1);

(b) graphs of circular chromatic number below three, at any degree; this is the circular memo's theorem [13, Corollary 3.2], re-derived here as a direct instance (Theorem 4.2);

(c) triangle-free graphs of maximum degree three, and every relation whose arcs join adjacent vertices of such a graph (*Theorem N3*, Theorem 4.3);

(d) loopless TT-free relations whose underlying graph has maximum degree three, directed triangles included (*Theorem U3*, Theorem 5.2);

(e) loopless TT-free relations whose vertices with at least three parents induce an acyclic subrelation, with no bound on in- or out-degrees (*Theorem T*, Theorem 5.3), and the version in which only parents inside a vertex's strongly connected component count (*Corollary* T_{int}, Corollary 5.4);

(f) loopless relations partitioned into blocks with private outside parents, each block K chordless subcubic or a chorded digon path, with $N[K]$ TT-free, or internally certified (*Theorem PC*, Theorem 5.10);

(g) every TT-free Cayley relation $\mathrm{Cay}(G, \{s, s^{-1}, t\})$, with arcs $g \to gx$, for every finite group G, when $\{s, s^{-1}, t\}$ has three elements and does not contain 1 (*Corollary CC*, Corollary 5.12).

Corollary T_{int} strictly contains the TT-free case of the two-input theorem of *Directed Recovery Structures* [15, Theorem 5.1], and Theorem U3 is the all-code version, on subcubic underlying graphs, of the fourth paper's scalar linear Theorem S [11, Theorem 6.13]. The certificates of Theorem PC come from *Lemma S* for chorded paths (Lemma 5.8) and from sweeps of chorded circulant cycles (*Proposition CC*, Proposition 5.9). The relation $\mathrm{Cay}(\mathbb{Z}_9, \{1, -1, 3\})$ lies in none of the classes (a)–(e); it is covered by (g), and by (f) with a single block.

Exact values (Section 6). On a triangle-free graph G of maximum degree three, every storage code satisfies $2\log|C| \le (|V| + |S| - |I(S)|)\log|\alpha|$ for every vertex set S, where $I(S)$ is the set of isolated vertices of $G - S$ (Theorem 6.1). With the fractional Tutte–Berge formula (Proposition 6.3), the guessing number equals the fractional matching number $\nu^*(G)$, attained over every alphabet of square size (Theorem 6.4 and Remark 6.5). This proves, for these graphs, Christofides and Markström's conjecture that fractional clique covers are optimal guessing strategies [19, 20], which is false in general [3, 18]; it was known before for perfect graphs, bipartite graphs in particular, and odd cycles [19], and for prisms and certain chorded cycles [27]. It also proves Dang's conjecture for planar triangle-free graphs at maximum degree three [20, Conjecture 15]. Dually, the broadcast rate of index coding is $|V| - \nu^*(G)$ (Theorem 6.7 and Remark 6.8). Relations that become one of our classes after deleting a set Z have storage codes of rate at most $(|V| + |Z|)/(2|V|)$ (Corollary 6.9).

Limits (Section 7). The natural rules for choosing pieces fail: $\mathrm{Cay}(\mathbb{Z}_9, \{1, -1, 3\})$ has no proper piece with private outside parents (Proposition 7.1); a 15-vertex relation with in- and out-degrees at most three has a unique minimum nonempty core, which is not a piece, although the relation has a global certificate (Remark 7.3); some chorded cycles have no valid sweep (Remark 7.4); and the Petersen graph has no global certificate, although Theorem N3 decomposes it into two pieces (Remark 7.5). The half bound fails on the folded cubes, whose odd girth equals their degree (Remark 7.6, from a rank formula of Barg, Schwartz and Yohananov [5]); so an exact half bound that assumes only $b_E(r) = 0$, for a fixed odd $r \ge 3$, cannot hold at maximum degree $r + 2$.

What this means for (T) (Section 8). These are unconditional storage-code theorems; they do not decide (T). They constrain constructions: in the cone relations of a correct oblivious transposer left by a genie of cost εL, no induced sub-relation on more than $2\varepsilon L$ coordinates may lie in any of the classes above. They also locate the remaining work for an impossibility proof: Conjecture 9.1, that every loopless TT-free relation of in-degree at most three has a decomposition, would give the degree-three case of the paired-distance question (Proposition 8.1).

1.3 How to read this paper: the formal proofs

Every numbered result of Sections 3 to 6 is formalized in Lean, with three exceptions: Corollary T_{int} is formalized in a block form, and the existence of a topological order of the strongly connected components, which turns the block form into the corollary, is proved in the text; the digon-component form of Theorem PC (Corollary 5.11) and the paid-deletion corollary (Corollary 6.9) have complete written proofs. Lemma 2.1, Proposition 7.1 and Proposition 8.1 are not formalized either and have complete proofs. Remarks 6.5 and 6.8, which pass from the formal code bounds of Theorems 6.4 and 6.7 to the guessing number and the broadcast rate, are labelled as not formalized and are derived in full in the text. The remarks of Section 7 report computations, reproducible with the scripts in `tools/`, and a cited rank formula for folded cubes; they are labelled as numerical or cited and are not used in proofs. The results of the fourth paper that we use (Theorem 2.2 and Lemmas 2.3 and 2.4) are Lean-verified there. Each numbered result states its status as described in Section 2.4.

1.4 Related work

Guessing numbers and graph entropy. Guessing games and guessing numbers were introduced by Riis [28]. Christofides and Markström [19] studied undirected graphs, introduced the fractional clique cover strategy and conjectured it to be optimal; the conjecture was disproved by Baber, Christofides, Dang, Vaughan and Riis [3], with triangle-free counterexamples by Cameron, Dang and Riis [18]. Dang's thesis [20] computes guessing numbers of several families of planar triangle-free graphs; those of maximum degree three are special cases of Theorem N3. Gadouleau and Riis [21] relate the guessing number of a digraph to the independence number of its guessing graph and construct digraphs with large guessing numbers. The bounds there come from Shannon and non-Shannon inequalities or from explicit strategies. Mazumdar, McGregor and Vorotnikova [27] bound storage capacity by covering a graph with gadgets of known capacity, which already gives combinatorial certificates for Shannon-type upper bounds on some families. Our certificates are of a different kind: elimination orders whose steps may use one outside vertex, with nested charges (Lemma E), composed by contraction rather than by covering (Theorem D).

Storage codes and index coding. Storage codes on graphs and their duality with index coding were studied by Mazumdar [26], by Shanmugam and Dimakis [30], and by Arbabjolfaei and Kim [1]. Index coding was introduced by Birk and Kol [7] and formalized, with minrank and the acyclic-subgraph lower bound, by Bar-Yossef, Birk, Jayram and Kol [4]; Blasiak, Kleinberg and Lubetzky [8] bound and approximate the broadcast rate by linear programming. Mazumdar, McGregor and Vorotnikova bounded the storage capacity above by vertex covers and below by fractional matchings [27]; the lower bound is Christofides and Markström's fractional clique cover strategy [19] in storage language, and on triangle-free subcubic graphs it is exact (Theorem 6.4). Without a degree bound, triangle-freeness does not force rate $\frac{1}{2}$: Barg and Zémor [6] and Barg, Schwartz and Yohananov [5] construct storage codes on triangle-free graphs of growing degree

with rates approaching $\frac{3}{4}$ and 1, and an earlier triangle-free family of rate approaching 1 is due to Golovnev and Haviv [22], as noted in [5].

The series. The fourth paper [11] proved that rate bounds pull back along homomorphisms (Theorem H), the half bound for scalar linear codes on triangle-free subcubic side information with its matching form (Theorem S), and that the Clebsch graph has guessing number ten. The circular memo [13] proved the exact partition identity on which Lemma E builds, and the half bound for circular chromatic number below three. The subcubic memo [17] proved half bounds for reversible and low-rank codes on subcubic supports and, for all codes, on triangle-free subcubic graphs covered by four-cycles; Theorem N3 contains the latter. *Directed Recovery Structures* [15] proved the paired-distance bound for cone relations, the two-input theorem and the hyperfinite half bound, and posed the all-code paired-distance question that motivates this paper. The covers paper [14] studies cover structure and local generation of linear storage codes. The companion paper [10], published together with this one, is about the depth side: its Theorem F proves depth reduction for head walks of bounded nesting load, so the crux alone, or an affirmative answer to Question 9.1 of *Directed Recovery Structures*, excludes oblivious linear-time transposers whose work heads enter every cell boundedly often.

To our knowledge, Lemma E, Theorems N3, U3, T and PC, and Corollary CC are new. Special cases were known: the guessing numbers of perfect graphs and odd cycles [19]; the storage capacity of prisms, of certain chorded cycles and of odd cycles [27]; the broadcast rates of cycles and of the cubic circulants, and numerically that of the Petersen graph [8]; the families of planar triangle-free graphs in Dang's thesis [20]; the half bound on square-covered subcubic graphs [17]; and the half bound for circular chromatic number below three, which is the circular memo's theorem [13].

1.5 Organization

Section 2 fixes notation. Section 3 proves Lemma E and Theorem D and the tools for building certificates. Section 4 treats 2-dicolourable relations and symmetric side information, Section 5 the directed classes, and Section 6 the barrier bound, guessing numbers and index codes. Section 7 shows where the method stops, Section 8 draws the consequences for constructions, and Section 9 lists open problems. The appendices index the Lean declarations and explain how to build and check the library.

2 Preliminaries

2.1 The statement (T)

We use the machine model of the first paper of this series [12]: a deterministic Turing machine with a read-only input tape, a constant number of work tapes over a finite alphabet, and a write-only output tape. For $d \geq 1$ let $b(d) = \max(1, \lceil \log_2 d \rceil)$. The input is a $d \times d$ matrix with $b(d)$-bit entries written row by row, of length $L = d^2 b(d)$, and the machine must write its transpose in the same encoding. Statement (T) says that some machine does this within $c(L + 1)$ steps, for a constant c and all d. Its formal version, `Transpose.LinearTimeTransposable`, is fixed in a frozen file of the first paper's library. A machine is *oblivious* if its head positions, output times and halting time depend only on the input length.

The Lean library of this paper does not model machines. Its theorems are about recovery relations, polymatroids, storage codes and index codes on finite sets. The connection to (T) goes

through the fourth paper of the series [11] and through *Directed Recovery Structures in Tape Computations* [15]; we recall it in Section 8.

2.2 Recovery relations

A *recovery relation* on a finite set V is a map $N\colon V \to 2^V$. We write $u \to v$, an *arc*, when $u \in N(v)$: then u is a *parent* (in-neighbour) of v and v a *child* of u. The *in-degree* of v is $|N(v)|$. The relation is *loopless* if $v \notin N(v)$ for every v. For $K \subseteq V$, the *induced relation* $N[K]$ on K is $v \mapsto N(v) \cap K$, and $N - Z = N[V \setminus Z]$. For a simple graph G on V, its *neighbourhood relation* N_G maps v to its set of neighbours (`MotionDepthCodes.nbhd`); every arc of N_G is reciprocated.

A *digon* is a pair of arcs $u \to v$ and $v \to u$; an arc that is not part of a digon is *one-way*. The *digon graph* $G_2(N)$ is the simple graph whose edges are the digons. The *underlying graph* $U(N)$ (`und`) joins $u \neq v$ whenever $u \to v$ or $v \to u$. A *transitive triangle* is a triple of distinct vertices x, y, z with $x \to y$, $y \to z$ and $x \to z$; we call y its *middle vertex*. A relation is *TT-free* if it has no transitive triangle (`TTFree`); `TTFreeOn` is the same condition for triples inside a set. In a loopless TT-free relation the parents of a vertex are pairwise non-adjacent, the digon graph is triangle-free, and every triangle of the underlying graph is a *directed* 3*-cycle* $x \to y \to z \to x$ without reverse arcs (Lemma 5.1 below); bidirected triangles are excluded.

A set $A \subseteq V$ is *acyclic* for N if every nonempty $T \subseteq A$ contains a vertex with no parent in T (`IsAcyclicIn`); equivalently, $N[A]$ has no loops and no directed cycles. A relation is 2*-dicolourable* if V is the union of two disjoint acyclic sets.

Paired incidence. *Directed Recovery Structures* [15, Definition 2.3] attaches to N its incidence graph $E(N)$: a bipartite graph with vertices L_p and R_p for $p \in V$ and an edge L_pR_q whenever $p \in N(q)$. It puts $b_E(r) = |\{p \in V : \mathrm{dist}_E(L_p, R_p) \leq r\}|$, and a sequence of relations satisfies the *paired-distance condition* if $b_E(r) = o(|V|)$ for every fixed r. Distances between L_p and R_p are odd. Our hypothesis "TT-free" is the exact form of this condition at radius three.

Lemma 2.1 (Transitive triangles are paired paths of length three)**.** Not formalized; a complete proof is given below.

Let N be a recovery relation and $p \in V$. Then $\mathrm{dist}_E(L_p, R_p) = 1$ if and only if $p \in N(p)$. If N is loopless, then $\mathrm{dist}_E(L_p, R_p) = 3$ if and only if p is the middle vertex of a transitive triangle. In particular, a loopless relation is TT-free if and only if $b_E(3) = 0$, that is, if and only if $\mathrm{dist}_E(L_p, R_p) \geq 5$ for every p.

Proof (not formalized). The edge L_pR_p exists exactly when $p \in N(p)$. Let N be loopless. A path L_p, R_z, L_x, R_p of length three means $p \in N(z)$, $x \in N(z)$ and $x \in N(p)$, that is, $p \to z$, $x \to z$ and $x \to p$. Its vertices are distinct, so $z \neq p$ and $x \neq p$, and $x \neq z$ because $x \in N(z)$ and N is loopless. Hence $x \to p \to z$ with $x \to z$ is a transitive triangle with middle vertex p. Conversely, every transitive triangle $x \to p \to z$, $x \to z$ gives this path. Since the distance is odd and not 1, it is at least 5 when no such path exists. □

2.3 Polymatroids and storage codes

We use the definitions of the fourth paper [11, Definitions 2.1 and 2.2]. A *polymatroid* on V is a function $h\colon 2^V \to \mathbb{R}$ with $h(\emptyset) = 0$ that is monotone and submodular. It is *valid* for N if $h(N(v) \cup \{v\}) = h(N(v))$ for every $v \in V$. We write $h(v)$ for $h(\{v\})$,

$$h(A \mid B) = h(A \cup B) - h(B), \qquad I(A; B \mid S) = h(A \cup S) + h(B \cup S) - h(A \cup B \cup S) - h(S) \geq 0,$$

and $I(w;X) = h(w) + h(X) - h(X \cup \{w\})$ (`mi`). The subject of this paper is the *weighted half bound*

$$2h(V) \;\le\; \sum_{v \in V} h(v) \qquad \text{for every polymatroid } h \text{ valid for } N. \tag{$\star$}$$

In the terms of the fourth paper, ($\star$) says that N satisfies the rate bound $\frac{1}{2}$; the formal statement is `MotionDepthCodes.RateBound` $N\,(1/2)$. A *storage code* for N over a finite alphabet α is a set $C \subseteq \alpha^V$ such that two words of C that agree on $N(v)$ agree at v, for every v (`MotionDepthCodes.IsStorageCode`). The *guessing number* $\mathrm{gn}(N,q)$ is the largest value of $\log_q|C|$ over storage codes with $|\alpha| = q$, and $\mathrm{gn}(N) = \sup_q \mathrm{gn}(N,q)$ is the asymptotic guessing number [28, 19].

Theorem 2.2 (Rate bounds bound codes)**.** Cited: the fourth paper [11, Theorem 6.3 and Corollary 6.9], Lean-verified there (`MotionDepthCodes.card_sq_le_of_rateBound_half`).

If N satisfies ($\star$), then every storage code $C \subseteq \alpha^V$ for N over a nonempty finite alphabet satisfies $|C|^2 \le |\alpha|^{|V|}$.

The proof applies ($\star$) to the entropy of a uniformly random codeword, which is a valid polymatroid. The same argument gives $|C|^2 \le \prod_v |\alpha_v|$ for codes with coordinate alphabets α_v [13, Corollary 3.4]; we do not use this.

Lemma 2.3 (Closure)**.** Cited: the fourth paper [11, Lemma 6.4], Lean-verified there (`MotionDepthCodes.IsPolymatroid.eq_of_closure`).

Let h be a polymatroid valid for N, and let $S \subseteq T$ be such that $N(v) \subseteq S$ for every $v \in T \setminus S$. Then $h(T) = h(S)$.

Lemma 2.4 (Fewer arcs)**.** Cited: the fourth paper's Theorem H [11, Theorem 6.5], applied to the identity map; Lean-verified there (`MotionDepthCodes.RateBound.of_hom`).

If $N(v) \subseteq N'(v)$ for every v and N' satisfies ($\star$), then so does N.

Indeed, a polymatroid valid for N is valid for N', by diminishing returns. The last tool is contraction.

Lemma 2.5 (Contraction)**.** Lean: `contr`, `contr_isPolymatroid`, `contr_insert_eq` [`Decomposition.lean`]

Let h be a polymatroid on V and $U \subseteq V$. Then $h_U(X) = h(X \cup U) - h(U)$ is a polymatroid. If h is valid for N and $N(v) \subseteq U \cup X$, then $h_U(X \cup \{v\}) = h_U(X)$; in particular the restriction of h_U to $V \setminus U$ is valid for $N - U$. Moreover $h_U(w) = h(w) - I(w;U)$ for $w \notin U$.

Outline of the formal proof. The polymatroid axioms for h_U follow from those for h, because $X \mapsto X \cup U$ preserves unions, intersections and inclusions. If $N(v) \subseteq U \cup X$, the closure lemma (Lemma 2.3) gives $h(X \cup U \cup \{v\}) = h(X \cup U)$. The last identity is the definition of $I(w;U)$. □

2.4 Formal status of the results

Every numbered result names its formal status in one of four ways. A line

Lean: `declaration, ...` [`file`]

gives the Lean declarations of a fully formalized result. The declarations live in the namespace `HalfRate` and the files in the directory `lean/HalfRate/`. The text after such a statement, labelled *Outline of the formal proof*, is a guide to the formal proof; it is a convenience, not a proof. A line beginning Lean (partial) marks a result whose formal counterpart covers all but an explicitly named

part, which the text proves. The words Not formalized mark a result with no formal counterpart; it comes with a complete proof, labelled *Proof (not formalized).* A line beginning Cited marks a result of an earlier paper of the series that is used without proof, and says where it is proved and whether it is Lean-verified there. Where the English statement simplifies, the Lean statement is authoritative. Numerical observations are labelled as such and are not used in proofs. Numbered definitions carry a status line only when they are formalized.

The library depends on Lean 4 and Mathlib (both v4.34.1), on the Lean library of the fourth paper (whose polymatroids, storage codes and rate bounds we use unchanged) and on that of its circular memo [13], imported through relative paths. It contains no `sorry`, no `native_decide` and no unsafe evaluation, and every declaration cited here depends only on the axioms `propext`, `Classical.choice` and `Quot.sound` (Appendix B). Appendix A lists every declaration of this library cited in the paper.

3 Nested elimination and the decomposition theorem

This section proves the engine of the paper. A *piece* H is eliminated one vertex at a time. When a vertex is eliminated it must be recovered from the vertices eliminated before it, from one *record* other than its own, and possibly from one vertex outside the piece. Lemma E bounds $2h(H)$ by the records plus the information that the outside vertices have about H, provided the steps charged to the same outside vertex are nested. Contracting H then reduces the half bound to the rest of the relation, and Theorem D composes pieces. Every other result of the paper is an instance: it only has to exhibit pieces.

3.1 Lemma E

Definition 3.1 (Elimination data)**.** Lean: `defect`, `totalDefect`, `Charged` [`NestedElimination.lean`]
Let $H \subseteq V$. *Elimination data* for H consist of finitely many *records* $P_i \subseteq H$, $i \in I$, not necessarily disjoint; an *elimination list* $v_0, \dots, v_{k-1}$, a listing of H without repetitions (the formal statement allows repetitions, which do not help); and maps $a, b\colon H \to I$ with $v \in P_{a(v)}$ and $b(v) \neq a(v)$ for every $v \in H$, the *own* and the *supplying* record of v. At step t the *revealed set* is $S_t = \{v_0, \dots, v_{t-1}\}$ and the *supply* is $Q_t = S_t \cup P_{b(v_t)}$. For a polymatroid g on V, the *defect* of step t is $g(v_t \mid Q_t)$.

Let $W \subseteq V$. A *charging* into W assigns to every vertex of H, and so to its step, either nothing or a vertex $w \in W$, such that the defect of a step with nothing vanishes (the step is *free*) and a step assigned to w satisfies $g(v_t \mid Q_t \cup \{w\}) = 0$ (it is *charged to* w). The charging is *nested* if, whenever $t' < t$ are consecutive steps charged to the same vertex, $Q_{t'} \cup \{v_{t'}\} \subseteq Q_t$.

Consecutive nesting implies $Q_{t'} \cup \{v_{t'}\} \subseteq Q_t$ for all steps $t' < t$ charged to the same vertex, since supplies grow along the chain.

Lemma 3.2 (Lemma E: nested elimination)**.** Lean: `record_increment`, `elim_defect`, `mi_mono`, `defect_eq_mi`, `totalDefect_le`, `nested_elimination` [`NestedElimination.lean`]
Let g be a polymatroid on V and fix elimination data for $H \subseteq V$.

(a) (Defect form.) $2g(H) \le \sum_i g(P_i) + \sum_t g(v_t \mid Q_t)$.

(b) (Charging.) If a charging into $W \subseteq V$ is nested, then $\sum_t g(v_t \mid Q_t) \le \sum_{w \in W} I(w; H)$.

(c) (Lemma E.) Consequently, if a charging into W is nested, then

$$2g(H) \;\le\; \sum_i g(P_i) + \sum_{w \in W} I(w; H).$$

Outline of the formal proof. (a) Let $m = |I|$; it is at least 2 as soon as $H \neq \emptyset$, since $a(v) \neq b(v)$. Induction along the list proves, for every t,

$$2g(H) + (m-2)\, g(S_t) \;\leq\; \sum_i g(S_t \cup P_i) + \sum_{s \geq t} g(v_s \mid Q_s).$$

At the end, $S_k = H$ contains every record and both sides equal $m\, g(H)$. In the inductive step $v = v_t$ is revealed: the own record does not change, because $v \in P_{a(v)}$; the supplying record grows by exactly the defect $g(v \mid S_t \cup P_{b(v)})$; and each of the other $m-2$ records grows by at most $g(S_t \cup \{v\}) - g(S_t)$, by submodularity (`record_increment`). At $t = 0$ this is (a).

(b) For $w \in W$ let $L(w)$ be the supply of the last step charged to w together with its vertex, initially $\emptyset$. Induction along the list proves $\sum_{s \geq t} g(v_s \mid Q_s) \leq \sum_{w \in W} (I(w; H) - I(w; L(w)))$, where L is taken at step t. A free step contributes nothing. For a step charged to w, the chain rule and $g(v_t \mid Q_t \cup \{w\}) = 0$ give $g(v_t \mid Q_t) = I(w; Q_t \cup \{v_t\}) - I(w; Q_t)$ (`defect_eq_mi`); nesting gives $L(w) \subseteq Q_t$, and $I(w; \cdot)$ is monotone (`mi_mono`), so the defect is at most $I(w; Q_t \cup \{v_t\}) - I(w; L(w))$, after which $L(w)$ becomes $Q_t \cup \{v_t\} \subseteq H$. At the start $I(w; \emptyset) = 0$.

(c) Combine (a) and (b). □

With $W = \emptyset$ and all steps free, part (c) is the inequality form of the exact partition identity of the circular memo [13, Theorem 2.1], which assumes that the records partition the ground set and identifies the slack exactly; part (a) is the inequality form of that memo's remark that dropping the supplying condition subtracts the defects. What is new is part (b): a step may use one vertex outside the piece, and several steps may use the same outside vertex, at the total price $I(w; H)$, provided their supplies are nested. Validity of g is not assumed, and the records need not be disjoint; validity enters in Theorem D through the closure lemma, and disjointness through subadditivity.

Why it matters. Lemma E is the only entropy inequality of the paper; everything after it is combinatorics. For (T) this means that the half-rate bounds needed by the impossibility programme of the fourth paper [11] can be proved by exhibiting finite combinatorial certificates. The charging of part (b) is what directed side information requires: in the directed classes of Section 5, an outside vertex can be a parent of several vertices of a piece. Not every natural piece admits such a charging; Remark 7.3 exhibits a minimum nonempty core that admits no piece certificate, with any number of records.

3.2 Piece certificates and Theorem D

Definition 3.3 (Piece certificates and decompositions)**.** Lean: `RelCharged`, `PieceCert`, `Decomp` [`Decomposition.lean`]

Let N be a recovery relation on V, let $U \subseteq V$ be a *context* and $H \subseteq V \setminus U$ a *piece*, and call the vertices of $V \setminus (U \cup H)$ *outside*. A *piece certificate* for H in context U consists of elimination data for H whose records $P_1, \ldots, P_m$ are pairwise disjoint with union H, and of an optional *outside parent* $p(v)$, an outside vertex, for each $v \in H$, such that for every step t:

(C1) $N(v_t) \subseteq U \cup Q_t \cup \{p(v_t)\}$, where $\{p(v_t)\} = \emptyset$ if v_t has no outside parent;

(C2) if $t' < t$ are consecutive steps with the same outside parent, then $Q_{t'} \cup \{v_{t'}\} \subseteq Q_t$.

So every parent of v_t outside the context has been eliminated, lies in the supplying record, or is the single outside parent of v_t. A *decomposition of N from context U* is a list of pieces $H_1, \ldots, H_k$ such that, with $U_j = U \cup H_1 \cup \cdots \cup H_{j-1}$, each H_j is disjoint from U_j and has a piece certificate in context U_j, and $U_{k+1} = V$. A *decomposition* of N is a decomposition from context $\emptyset$.

Theorem 3.4 (Theorem D: decomposition)**.** Lean: `contr_isPolymatroid`, `RelCharged.charged`, `PieceCert.ineq`, `decomp_ineq`, `rateBound_of_decomp`, `card_sq_le_of_decomp` [`Decomposition.lean`]

Let h be a polymatroid valid for N.

(a) (Pieces.) If H has a piece certificate in context U, then, with the contraction h_U of Lemma 2.5 and the information I_{h_U} computed in h_U,

$$2h_U(H) \;\le\; \sum_{v\in H} h_U(v) + \sum_{w\notin U\cup H} I_{h_U}(w;H).$$

(b) (Relative form.) If N has a decomposition from context U, then

$$2\big(h(V)-h(U)\big) \;\le\; \sum_{v\notin U} \big(h(U\cup\{v\})-h(U)\big).$$

(c) (Theorem D.) If N has a decomposition, then N satisfies $(\star)$, and every storage code $C\subseteq \alpha^V$ for N over a nonempty finite alphabet satisfies $|C|^2 \le |\alpha|^{|V|}$.

Outline of the formal proof. (a) Apply Lemma E to $g = h_U$, a polymatroid by Lemma 2.5, with W the set of outside vertices. Condition (C1) and Lemma 2.5 turn each step into a free step or a step charged to its outside parent (`RelCharged.charged`), and (C2) is the nesting. Finally $\sum_i h_U(P_i) \le \sum_{v\in H} h_U(v)$ by subadditivity over the disjoint records.

(b) Induction over the list of pieces. For the first piece H in context U, part (a) applies; the induction hypothesis, for the remaining pieces from context $U\cup H$, gives $2(h(V)-h(U\cup H)) \le \sum_{w\notin U\cup H}(h(U\cup H\cup\{w\})-h(U\cup H))$. Since $h(U\cup H\cup\{w\})-h(U\cup H) = h_U(w) - I_{h_U}(w;H)$, the information terms cancel when the two inequalities are added, and $h(V)-h(U) = h_U(H) + (h(V)-h(U\cup H))$.

(c) Take $U=\emptyset$ in (b); the code bound is Theorem 2.2. □

The pieces are certified in the relation that remains after contracting the earlier pieces: parents in earlier pieces are free. The bookkeeping is exact, in the sense that the piece inequalities add up to $\sum_v h(v) - 2h(V)$ with all outside charges cancelling; unfolding a decomposition therefore gives an explicit Shannon certificate for $(\star)$, a nonnegative combination of submodularity and validity brackets.

Why it matters. Theorem D turns each class theorem below into a statement that decompositions exist, and it is the only route by which this paper proves half-rate bounds. For (T), a proof of the fourth paper's crux along these lines would have to decompose the cone relations of oblivious schedules into certified pieces; Section 7 shows where the natural rules for choosing pieces break down.

3.3 Building certificates

Three tools produce piece certificates. The first removes the need to check nesting.

Lemma 3.5 (Co-supply implies nesting)**.** Lean: `rev`, `relCharged_of_cosupply` [`Positional.lean`]

Let elimination data for H, with records partitioning H and optional outside parents, satisfy (C1) at every step. If all steps with the same outside parent have the same supplying record (the outside parents are co-supplied*), then (C2) holds. In particular (C2) holds if every outside vertex is the outside parent of at most one vertex of H (the outside parents are* private*).*

Outline of the formal proof. For steps $t' < t$ with the common supplying record P_b, $Q_{t'} \cup \{v_{t'}\} = S_{t'+1} \cup P_b \subseteq S_t \cup P_b = Q_t$. The formal statement takes the step conditions position by position along the list, with `rev` the revealed set before a position. □

Lemma 3.6 (Fewer arcs and greedy certificates)**.** Lean: `RelCharged.mono`, `PieceCert.mono`, `exists_relCharged`, `pieceCert_of_elim` [`Pieces.lean`]

Let $U \subseteq V$ and $H \subseteq V \setminus U$.

(a) If $N(v) \subseteq N'(v)$ for every $v \in H$, every piece certificate for H in context U for N' is one for N.

(b) Let records $P_1, \dots, P_m$ partition H, with own and supplying records $a(v) \neq b(v)$ and optional co-supplied outside parents $p(v)$. If every proper subset $S \subsetneq H$ misses a vertex $v \in H$ with $N(v) \subseteq U \cup S \cup P_{b(v)} \cup \{p(v)\}$, then H has a piece certificate in context U.

Outline of the formal proof. (a) Condition (C1) only becomes weaker, and (C2) does not mention N. (b) Starting from $S = \emptyset$, repeatedly eliminate a vertex given by the hypothesis. Each step satisfies (C1) by construction, and co-supply gives (C2) (Lemma 3.5). The list ends when $S = H$. □

Lemma 3.7 (Lemma B_{dir}: relative 2-dicolourings)**.** Lean: `bdirPiece` [`Pieces.lean`]

Let $U \subseteq V$ and let $H = A \sqcup B \subseteq V \setminus U$, where A and B are acyclic for N. Suppose that every vertex of H has at most one parent outside $U \cup H$, and that every outside vertex has all its children in H inside A, or all inside B. Then H has a piece certificate in context U (the formal proof uses the two records A and B).

Outline of the formal proof. The supplying record of a vertex of A is B, and that of a vertex of B is A; the outside parent of a vertex is its parent outside $U \cup H$, if any. Apply Lemma 3.6(b). Given a proper subset S of H: if $A \not\subseteq S$, acyclicity gives a vertex of $A \setminus S$ with no parent in $A \setminus S$, whose parents in H are therefore revealed or in B; otherwise $A \subseteq S$, and a vertex of $B \setminus S$ with no parent in $B \setminus S$ has all its parents in H revealed. Co-supply holds because the children of an outside vertex lie in one part, which has one supplying record. □

Why it matters. Relative 2-dicolourings are the workhorse of Section 5: directed triangles, shortest directed cycles of in-degree-two vertices and the final step of Theorem T are pieces of this kind. They need no degree bound and allow an outside vertex to have many children in a piece, as long as these lie on one side.

3.4 Global certificates and pull-back

A piece certificate with $H = V$ has no outside vertices. Its order-free form is the following.

Definition 3.8 (Global elimination certificates)**.** Lean: `ElimCert` [`Corollaries.lean`]

A *global elimination certificate* for N consists of records $P_1, \dots, P_m$ partitioning V and own and supplying records $a(v) \neq b(v)$ with $v \in P_{a(v)}$, such that every proper subset $S \subsetneq V$ misses a vertex v with $N(v) \subseteq S \cup P_{b(v)}$.

Equivalently, there is an elimination list in which the parents of each vertex that are not earlier in the list lie in its supplying record: given the list and a proper subset S, the first vertex of the list outside S is eligible.

Proposition 3.9 (Global certificates)**.** Lean: `ElimCert.rateBound` [`Corollaries.lean`]
Lean: `sum_records_step` [`NestedElimination.lean`]

If N has a global elimination certificate, then N satisfies ($\star$).

Outline of the formal proof. Downward induction on $|V \setminus S|$ proves $2h(V)+(m-2)h(S) \le \sum_i h(S \cup P_i)$, as in Lemma 3.2(a), with every defect zero by validity and the closure lemma; the inductive step is `sum_records_step`. At $S = \emptyset$, subadditivity over the records gives ($\star$). □

With three records and cyclic supply, $b(v)$ the record after $a(v)$, this is the three-record criterion of the circular memo [13, Lemma 2.2] for partitions; the memo also allows overlapping records. Global certificates pull back along homomorphisms of recovery relations.

Lemma 3.10 (Pull-back)**.** Lean: `ElimCert.pullback` [`Corollaries.lean`]

Let N and N' be recovery relations on V and W, and let $\varphi\colon V \to W$ satisfy $\varphi(u) \in N'(\varphi(v))$ whenever $u \in N(v)$. If N' has a global elimination certificate with records P'_i and maps a', b', then N has one with records $\varphi^{-1}(P'_i)$ and maps $a' \circ \varphi$, $b' \circ \varphi$.

Outline of the formal proof. Given a proper subset $S \subsetneq V$, let S' be the set of $x \in W$ whose whole fibre $\varphi^{-1}(x)$ lies in S. It is a proper subset of W, so some $x \notin S'$ satisfies $N'(x) \subseteq S' \cup P'_{b'(x)}$; choose $v \in \varphi^{-1}(x) \setminus S$. A parent u of v has $\varphi(u) \in N'(x)$: either $\varphi(u) \in S'$, so $u \in S$, or $\varphi(u) \in P'_{b'(x)}$, so u lies in the supplying record of v. □

No looplessness is needed: a relation with a global certificate is loopless, since a loop at v would put v in its own supplying record. The fourth paper's Theorem H [11, Theorem 6.5] pushes polymatroids forward along homomorphisms and so transfers ($\star$) itself; Lemma 3.10 transfers the certificate instead, and needs no polymatroid argument.

Why it matters. One certificate on a small target proves ($\star$) for every relation that maps to it, whatever its size or degree. For (T), an impossibility proof could map the cone relations of a schedule to certified targets; a construction must avoid side information that admits such a map. Global certificates are weaker than decompositions (Remark 7.5), which is why the classes of Sections 4 and 5 use Theorem D.

4 Base cases and symmetric side information

This section treats the base case of 2-dicolourable relations and two classes of graphs: graphs of circular chromatic number below three, at any degree, and triangle-free graphs of maximum degree three.

Proposition 4.1 (2-dicolourable relations)**.** Lean: `IsAcyclicIn`, `rateBound_of_twoAcyclic` [`Corollaries.lean`]

If V is the disjoint union of two sets A and B that are acyclic for N, then N satisfies ($\star$).

Outline of the formal proof. A global elimination certificate with records A and B, each the supplying record of the other: while A is not revealed, eliminate a vertex of the unrevealed part of A with no parent there; then do the same in B. Apply Proposition 3.9. □

Bipartite graphs are examples, and so is every loopless relation whose underlying graph is a forest: its directed cycles are digons, so a proper 2-colouring of the underlying graph is a 2-dicolouring. The bound is classical: for a valid h, each of A and B determines the other in topological order, so $h(V) = h(A) = h(B)$.

Why it matters. This is the base case of every decomposition below: pieces end, after their cyclic parts have been removed, in 2-dicolourings. For (T) it says that side information without odd dependency structure, in this directed sense, gives no coding gain at all.

4.1 Circular chromatic number below three

The *circular clique* $K_{p/q}$ has vertices $\mathbb{Z}/p$, with $i \sim j$ if and only if $q \le (j-i) \bmod p \le p-q$. The *Andrásfai graph* $\mathrm{And}(k)$, $k \ge 2$, is $K_{(3k-1)/k}$. The circular chromatic number $\chi_c(G)$ of a finite graph is the least p/q such that G maps homomorphically to $K_{p/q}$ [31, 32]; so $\chi_c(G) < 3$ if and only if G maps to some $K_{p/q}$ with $p < 3q$.

Theorem 4.2 (Circular chromatic number below three)**.** Lean: `andrasfaiCert`, `rateBound_of_circularHom'` [`Corollaries.lean`]

Let G be a finite simple graph with a homomorphism to a circular clique $K_{p/q}$ with $q > 0$ and $p < 3q$, that is, with $\chi_c(G) < 3$. Then N_G satisfies ($\star$).

This is the circular memo's theorem [13, Corollary 3.2], Lean-verified there through the fourth paper's Theorem H. We re-derive it as a direct instance of global certificates.

Outline of the formal proof. The Andrásfai graph $\mathrm{And}(k)$ has a global elimination certificate with three records (`andrasfaiCert`): the intervals $[0,k)$, $[k,2k)$ and $[2k,3k-1)$, each supplied by the next one cyclically. The elimination order is the circular memo's interleaved order, with keys $3i$, $3j+1$ and $3\ell+2$ for the i-th, j-th and ℓ-th vertex of the three intervals, and the memo's adjacency lemma (`CircularMemo.neighbor_order`) says that every later neighbour of a vertex lies in the next interval. For $k = q+1$ we have $pk \le (3k-1)q$, so the Bondy–Hell map $i \mapsto \lfloor (3k-1)i/p \rfloor$ is a homomorphism $K_{p/q} \to \mathrm{And}(k)$ [9], formalized in the fourth paper (`MotionDepthCodes.circularCliqueHom`). Compose it with the given homomorphism, pull the certificate back (Lemma 3.10) and apply Proposition 3.9. □

Why it matters. The theorem has no degree assumption: every graph of circular chromatic number below three, for instance every odd cycle of length at least five and every graph mapping to one, satisfies the half bound for all codes over all alphabets. For (T), symmetric side information of this kind cannot carry a coding gain however dense it is. The bound does not reach $\chi_c = 3$; the Petersen graph, which has $\chi_c = 3$ [25], is covered by the fourth paper's certificate [11, Corollary 6.9] and by the next theorem.

4.2 Triangle-free graphs of maximum degree three

Theorem 4.3 (Theorem N3)**.** Lean: `CycIn`, `CycIn.chordless`, `CycIn.share`, `cyclePiece`, `decomp_of_cycles`, `n3_decomp`, `rateBound_n3`, `card_sq_le_n3` [`N3.lean`]

Let G be a triangle-free graph of maximum degree at most 3. Then N_G has a decomposition. Hence N_G satisfies ($\star$), and every storage code C for G over a nonempty finite alphabet satisfies $|C|^2 \le |\alpha|^{|V|}$.

Outline of the formal proof. By strong induction on the set R of remaining vertices, with context $V \setminus R$, `decomp_of_cycles` constructs a decomposition from every context. If $G[R]$ contains a cycle, let $C = c_0 c_1 \cdots c_{g-1}$ be a shortest one (`CycIn`); $g \ge 4$ because G is triangle-free. Three facts:

(i) C is chordless (`CycIn.chordless`), since a chord would close a shorter cycle;

(ii) if a vertex $w \in R$ off C is adjacent to c_i and c_j, $i < j$, it closes cycles of lengths $j - i + 2$ and $g - (j - i) + 2$, both at least g; hence $g = 4$ and c_i, c_j are opposite (`CycIn.share`);

(iii) every c_i has at most one neighbour off C, since it has two on C and degree at most three.

The piece certificate of C in context $V \setminus R$ (`cyclePiece`) eliminates $c_0, \ldots, c_{g-1}$ in order with three records given by colours: c_0 has colour 0 and c_{g-1} colour 1; if g is odd, c_{g-2} has colour 2; every other c_i has colour 1 for odd i and 0 for even i. The supplying record of c_t is the colour of c_{t+1}, and that of c_{g-1} is colour 0. At a step $t \geq 1$ the cycle neighbours of c_t are c_{t-1}, already eliminated, and c_{t+1}, in the supplying record; at step 0 they are c_1 and c_{g-1}, both of colour 1. The remaining neighbour of c_t, if it lies in $R \setminus C$, is its outside parent. By (ii) outside parents are private, except for opposite vertices of a 4-cycle, which have equal supplying records (c_0, c_2 are supplied by colour 1, and c_1, c_3 by colour 0); Lemma 3.5 applies. If $G[R]$ is a forest, a proper 2-colouring of it is a piece by Lemma 3.7. Theorem D concludes. □

By Lemma 2.4, the half bound holds also for every relation N with $N(v) \subseteq N_G(v)$ for all v, that is, for every relation whose arcs join adjacent vertices of G, in one direction or in both; `decomp_of_cycles` gives decompositions for such relations as well, its hypothesis on 3-cycles being vacuous here. Several known results are special cases.

- The fourth paper's Theorem S [11, Theorem 6.13] proves the half bound for scalar linear codes, over every field, on a directed class that contains these graphs; Theorem N3 is its all-code version on triangle-free subcubic graphs. For scalar linear codes the exact value is $|V| - \nu(G)$ for the minrank, with the integral matching number ν [11, Theorem 6.16]; for all codes the guessing number is the fractional matching number $\nu^*(G)$ (Theorem 6.4). The two differ already on C_5, where $\nu = 2$ and $\nu^* = 5/2$.
- The subcubic memo [17, Theorem 7.1] proves the half bound, in Lean, for triangle-free subcubic graphs all of whose edges lie in four-cycles, through a homomorphism to $K_{8/3}$; this is a special case. Its information-set theorem [17, Theorem 2.3] bounds codes that are reversible on the rows by $q^{\nu(G)}$, which is stronger on that class of codes.
- The circular memo's theorem (Theorem 4.2) covers graphs of every degree with $\chi_c < 3$, and the fourth paper's certificates cover the graphs that map to the Petersen graph [11, Corollary 6.9]. Theorem N3 needs no target: it covers every triangle-free subcubic graph. In particular it proves unconditionally the fourth paper's conditional result for cubic graphs of large girth [11, Corollary 6.10], which assumed that such graphs map to C_5 or to $K_{14/5}$.
- Dang [20] computes the guessing numbers of several families of planar triangle-free graphs and conjectures that the fractional clique cover strategy is optimal on every planar triangle-free graph [20, Conjecture 15]. With Theorem 6.4 below, Theorem N3 proves this for maximum degree three, without planarity.

Why it matters. No storage code on a triangle-free subcubic graph beats the pairing, whatever its alphabet and whether or not it is linear. With Corollary 6.9, this answers the fourth paper's undirected crux [11, Question 6.1] affirmatively at maximum degree three: in a locally bipartite family of maximum degree at most three, only $o(|V|)$ vertices lie on triangles, deleting them leaves a triangle-free subcubic graph, and so every storage code has rate at most $\frac{1}{2} + o(1)$. For (T), symmetric subcubic side information that is triangle-free cannot carry a coding gain. Native cone relations are directed, however, and their symmetrizations can contain linearly many triangles [15, Proposition 3.10]; this is why Section 5 allows directed triangles.

5 Directed relations without transitive triangles

By Lemma 2.1, a loopless relation is TT-free exactly when every coordinate has paired distance at least five. This section proves the half bound for three classes of such relations: subcubic underlying graphs (Theorem U3), relations whose vertices of in-degree at least three are acyclic (Theorem T), and block structures with private outside parents (Theorem PC), which contain all TT-free Cayley relations $\mathrm{Cay}(G, \{s, s^{-1}, t\})$ (Corollary CC). Directed triangles are allowed throughout; they are the only triangles a TT-free relation has.

Lemma 5.1 (Triangles of TT-free relations)**.** Lean: `dtri_orient` [U3.lean]

Let N be TT-free, and let x_0, x_1, x_2 be distinct and pairwise adjacent in the underlying graph, with $x_0 \to x_1$. Then $x_1 \to x_2$ and $x_2 \to x_0$, and none of the arcs $x_1 \to x_0$, $x_2 \to x_1$, $x_0 \to x_2$ is present.

Outline of the formal proof. If $x_2 \to x_1$, then either $x_0 \to x_2$, and $x_0 \to x_2 \to x_1$ with $x_0 \to x_1$ is a transitive triangle, or $x_2 \to x_0$, and $x_2 \to x_0 \to x_1$ with $x_2 \to x_1$ is one. So $x_1 \to x_2$, and then $x_0 \to x_2$ would close the transitive triangle $x_0 \to x_1 \to x_2$; hence $x_2 \to x_0$. The reverse arcs $x_1 \to x_0$ and $x_2 \to x_1$ would close the transitive triangles $x_1 \to x_2 \to x_0$, $x_1 \to x_0$ and $x_2 \to x_0 \to x_1$, $x_2 \to x_1$. □

5.1 Subcubic underlying graphs: Theorem U3

Theorem 5.2 (Theorem U3)**.** Lean: `und`, `sub_nbhd_und`, `dtriPiece`, `triPiece`, `rateBound_u3`, `card_sq_le_u3` [U3.lean]

Let N be loopless and TT-free, and let its underlying graph have maximum degree at most 3. Then N satisfies ($\star$), and every storage code C for N over a nonempty finite alphabet satisfies $|C|^2 \le |\alpha|^{|V|}$.

Outline of the formal proof. Let $G = U(N)$; since N is loopless, $N(v) \subseteq N_G(v)$ (`sub_nbhd_und`). Run the shortest-cycle decomposition of Theorem N3 on G (`decomp_of_cycles`). Shortest cycles of length at least four are pieces for N_G and hence for N, which has fewer arcs (Lemma 3.6(a)); the acyclic remainder is a relative 2-colouring, built directly for N by Lemma 3.7. The new case is a shortest cycle of length three. By Lemma 5.1 it is a directed 3-cycle $c_0 \to c_1 \to c_2 \to c_0$ without reverse arcs (`triPiece`). Take $A = \{c_0, c_2\}$, which carries the single arc $c_2 \to c_0$ and is acyclic, and $B = \{c_1\}$ (`dtriPiece`). Every c_i has two neighbours on the triangle, hence at most one parent outside it. Two distinct triangle vertices cannot share an outside parent w: with the arc between them, w would complete a transitive triangle. So outside parents are private, and Lemma 3.7 applies. □

When the underlying graph is triangle-free, Theorem U3 follows from Theorem N3 and Lemma 2.4; its new content is directed triangles. The fourth paper's Theorem S [11, Theorem 6.13] proves the half bound for scalar linear codes on a larger class: its condition "no recovery triangle" is TT-freeness, and its reciprocity condition asks that a vertex with three parents have no child that is not a parent, which holds when the underlying graph is subcubic. Theorem U3 is thus the all-code version of Theorem S on relations with subcubic underlying graph; the all-code version of Theorem S on its whole class is open.

Why it matters. Directed triangles do not help a code. This matters for (T) because native cone relations can contain linearly many disjoint directed 3-cycles [15, Proposition 3.10], which rule out the symmetric hypothesis of Theorem N3 but not the hypotheses of this section.

5.2 Acyclic high in-degree: Theorem T

Theorem 5.3 (Theorem T)**.** Lean: `DCyc`, `exists_hasDCyc_of_sourceless`, `DCyc.chordless`, `DCyc.acyclic_of_missing`, `singlePiece`, `i5Piece`, `block_decomp`, `rateBound_thmT`, `card_sq_le_thmT` [`TheoremT.lean`]

Let N be loopless and TT-free, and suppose that the set $\{v : |N(v)| \geq 3\}$ of vertices with at least three parents is acyclic for N. Then N satisfies ($\star$), and every storage code C for N over a nonempty finite alphabet satisfies $|C|^2 \leq |\alpha|^{|V|}$. No bound on in- or out-degrees is assumed.

Outline of the formal proof. By strong induction on the set R of remaining vertices (`block_decomp`, with one block), in-degrees being counted inside R:

(a) If some $v \in R$ has at most one parent in R, then $\{v\}$ is a piece, its remaining parent being its outside parent (`singlePiece`, Lemma 3.7 with $A = \{v\}$ and $B = \emptyset$).

(b) Otherwise let W_R be the set of vertices with exactly two parents in R. If W_R is not acyclic, it contains a nonempty set in which every vertex has a parent, hence a directed cycle (`exists_hasDCyc_of_sourceless`). A shortest directed cycle C in W_R (`DCyc`) is induced (`DCyc.chordless`): the only parent of each $c \in C$ on C is its predecessor. Hence each $c \in C$ has exactly one further parent in R, off C, its outside parent $p(c)$. Fix $c_0 \in C$ and let $A = \{c \in C : p(c) = p(c_0)\}$ and $B = C \setminus A$. If $c_0 \to c_1$ on C, then $p(c_1) \neq p(c_0)$, since otherwise $p(c_0) \to c_0 \to c_1$ with $p(c_0) \to c_1$ is a transitive triangle. So A and B are proper subsets of an induced cycle and hence acyclic (`DCyc.acyclic_of_missing`), and every outside vertex has its children on C inside A (if it is $p(c_0)$) or inside B. Lemma 3.7 applies (`i5Piece`).

(c) Otherwise W_R is acyclic, and so is the set of vertices with at least three parents in R, a subset of the acyclic set of the hypothesis. These two sets partition R, which is a piece by Lemma 3.7.

Theorem D concludes. □

Strongly connected components can be treated separately, since a parent in an earlier component can be contracted. The formal statement is a block form.

Corollary 5.4 (Corollary T_{int})**.** Lean (partial): `TTFreeOn`, `BlockOrder`, `blocks_decomp`, `rateBound_blocks` [`TheoremT.lean`]

Not formalized: the existence of a topological order of the strongly connected components, which is standard; it is proved below.

Let N be loopless. Suppose that no strongly connected component of N contains a transitive triangle, and that the set T_{int} of vertices with at least three parents inside their own strongly connected component is acyclic. Then N satisfies ($\star$). There is no restriction on parents outside a vertex's component.

The formal theorem (`rateBound_blocks`) is the following block form: if N is loopless and V is the union of pairwise disjoint blocks $S_1, \ldots, S_r$ such that every parent of a vertex of S_j lies in $S_1 \cup \cdots \cup S_j$, no block contains a transitive triangle, and each set $\{v \in S_j : |N(v) \cap S_j| \geq 3\}$ is acyclic, then N satisfies ($\star$).

Outline of the formal proof. Induction over the list of blocks (`blocks_decomp`): each block S_j is decomposed by the procedure of Theorem T run inside S_j (`block_decomp`), with the earlier blocks in the context. Parents in earlier blocks are then free, so in-degrees in (a)–(c) are counted inside the remaining part of S_j, and TT-freeness is needed only inside S_j (`TTFreeOn`). □

Proof (not formalized). We derive the corollary from the block form. The *condensation* of N has the strongly connected components as vertices and an arc $S \to S'$, $S \neq S'$, whenever some arc of N goes from S to S'. It has no directed cycle, since components on such a cycle would be strongly connected to each other. Hence it has a topological order: repeatedly remove a component that receives no arc from the remaining ones. Listing the components $S_1, \dots, S_r$ in this order, every parent of a vertex of S_j lies in $S_1 \cup \cdots \cup S_j$. The set $\{v \in S_j : |N(v) \cap S_j| \geq 3\}$ is $T_{\mathrm{int}} \cap S_j$, which is acyclic because acyclicity passes to subsets. The block form applies. □

When $T_{\mathrm{int}} = \emptyset$, that is, when every vertex has at most two parents inside its component, Corollary T_{int} is the TT-free case of the two-input theorem of *Directed Recovery Structures* [15, Theorem 5.1]: TT-free components contain no bidirected triangles, so that theorem's correction term vanishes. The corollary strictly contains this case, because it allows vertices with three or more internal parents as long as they induce an acyclic sub-relation. Its proof dismantles a bidirected odd cycle by contracting one digon, a shortest directed cycle of in-degree-two vertices, and does not use the directed Brooks theorem on which the proof in [15] relies. Conversely, [15, Theorem 5.1] also covers components that contain transitive triangles, which we do not.

Why it matters. Theorem T and Corollary T_{int} have no degree bound. For (T) they locate where a coding gain on TT-free side information must come from: from directed cycles through vertices that have at least three parents in their own component. In a cone relation, after the middle vertices of transitive triangles have been deleted at the price of Corollary 6.9, a construction needs such cycles on a large fraction of the coordinates (Section 8).

5.3 Blocks with private outside parents: Theorem PC

The pieces of Theorems N3, U3 and T are small. The next theorem takes whole blocks as pieces, in any order; a block then needs only an internal certificate.

Definition 5.5 (Elimination certificates of a block)**.** Lean: `ElimOn` [`Blocks.lean`]

Let $K \subseteq V$. An *elimination certificate* of $N[K]$ consists of an elimination list of K without repetitions, records partitioning K, and own and supplying records $a(v) \neq b(v)$ with $v \in P_{a(v)}$, such that at every step t the parents of v_t inside K lie in $S_t \cup P_{b(v_t)}$.

This is a global elimination certificate of the induced relation $N[K]$ (Definition 3.8), presented by its list.

Lemma 5.6 (Internal certificates give pieces)**.** Lean: `ElimOn.piece` [`Blocks.lean`]

Let $N[K]$ have an elimination certificate. Suppose that every vertex of K has at most one parent outside K, and that every vertex outside K is a parent of at most one vertex of K. Then K has a piece certificate in every context U disjoint from K.

Outline of the formal proof. Keep the list and the records. The outside parent of $v \in K$ is its parent outside $U \cup K$, if any; its other parents lie in U or in K, where the internal certificate applies. Outside parents are private, so Lemma 3.5 gives nesting. (The formal statement asks for at most one parent in $V \setminus (U \cup K)$ only.) □

Lemma 5.7 (Positional certificates; chordless digon cycles)**.** Lean: `elimOn_ofFn`, `elimOn_cycle` [`Circulant.lean`]

Let $K = \{k_0, \dots, k_{m-1}\}$ be nonempty and listed without repetitions.

(a) Let col *and* bcol *be colourings of the positions* $0, \dots, m-1$ *with* $\mathrm{col}(j) \neq \mathrm{bcol}(j)$ *for every* j*. If* $k_i \to k_j$ *implies* $i < j$ *or* $\mathrm{col}(i) = \mathrm{bcol}(j)$*, then* $N[K]$ *has an elimination certificate (the formal proof uses the list* $k_0, \dots, k_{m-1}$ *and the colour classes as records).*

(b) If $m \geq 4$ *and every arc inside* K *joins cyclically consecutive vertices (*$k_i \to k_j$ *only if* $i \equiv j \pm 1 \pmod m$*), then* $N[K]$ *has an elimination certificate.*

Outline of the formal proof. (a) The own record of k_j is its colour and its supplying record is $\mathrm{bcol}(j)$; a parent k_i that comes later has colour $\mathrm{bcol}(j)$. (b) Use the colouring of the cycle pieces of Theorem N3, with $\mathrm{bcol}(j) = \mathrm{col}(j+1)$ for $j < m-1$ and $\mathrm{bcol}(m-1) = 0$. □

Along a list $k_0, \dots, k_{L-1}$, call an arc $k_u \to k_v$ with $u \geq v+3$ a *downward chord*.

Lemma 5.8 (Lemma S: chorded paths)**.** Lean: `DownChord`, `pathRoot`, `pathRoot_succ_ne`, `elimOn_path` [`PathCert.lean`]

Let $K = \{k_0, \dots, k_{L-1}\}$ *be nonempty and listed without repetitions. Suppose that every arc* $k_i \to k_j$ *inside* K *joins consecutive positions,* $|i-j| = 1$*, or skips at least two positions,* $|i-j| \geq 3$*, and that every vertex of* K *is the head of at most one downward chord and the tail of at most one. Then* $N[K]$ *has an elimination certificate (the formal proof uses the list* $k_0, \dots, k_{L-1}$ *and, as records, the classes of the equivalence relation on positions generated by the* links $u \sim v+1$*, one for each downward chord* $k_u \to k_v$*, together with one empty record).*

Outline of the formal proof. When k_j is eliminated, its parents inside K that come later are k_{j+1} (if $j < L-1$) and the tail k_u of the downward chord to k_j, if there is one; these two are linked, so they lie in one class, which serves as supplying record (through Lemma 5.7(a)). It remains to show that j and $j+1$ are never in the same class (`pathRoot_succ_ne`). A link joins u to $v+1 \leq u-2$. A position is the upper end of at most one link, from the downward chord out of it, and the lower end of at most one link, from the downward chord into the position below it, because these chords are unique. So the classes are chains whose positions decrease by at least two at each link: following links downwards from any position reaches a lowest position, its *root* (`pathRoot`), and two positions of one class differ by at least two. □

A *chorded digon path* is a set $K = \{k_0, \dots, k_{L-1}\}$ such that $k_i \leftrightarrow k_{i+1}$ is a digon for $0 \leq i < L-1$ and the other arcs inside K, the *chords*, number at most one into and at most one out of every vertex. If N is loopless and $N[K]$ is TT-free, a chorded digon path satisfies the hypotheses of Lemma S (`ChordedPath.elimOn`): a chord between k_i and $k_{i\pm2}$ would close a transitive triangle with the two digons around $k_{i\pm1}$.

Proposition 5.9 (Proposition CC: chorded circulant cycles)**.** Lean: `circCol`, `circB`, `elimOn_circulant`, `circulant_sweep` [`Circulant.lean`]

Let $3 \leq k \leq m-3$ *and let* $K = \{k_0, \dots, k_{m-1}\}$ *be listed without repetitions so that every arc inside* K *has the form* $k_{j\pm1} \to k_j$ *or* $k_{j-k} \to k_j$*, indices modulo* m*. Put* $\delta = m-1-k$*.*

(a) If δ *divides neither* $m-1$ *nor* $m-3$*, then* $N[K]$ *has an elimination certificate (the formal proof uses the sweep along* $k_0, k_1, \dots, k_{m-1}$ *described below).*

(b) If $2k \not\equiv \pm1 \pmod m$*, then the condition of (a) holds for the offset* k *or for the offset* $m-k$*.*

Outline of the formal proof. (a) (`elimOn_circulant`, through Lemma 5.7(a).) Position 0 has a record of its own; a position $p \neq 0$ gets the record of its residue modulo δ, except that the residue class of $m-1$ is merged into that of 1 (`circCol`). The supplying record of $p < m-1$ is the record

of $p+1$, and that of $m-1$ is the record of position 0 (`circB`). The later parents of k_p are k_{p+1}; for $p=0$ also k_{m-1}, which has the merged colour of k_1; and, for $p<k$, the tail k_{p-k+m} of the chord, whose position $p-k+m=(p+1)+\delta$ is congruent to $p+1$. Own and supplying records differ: for $1\le p<m-1$ the residues of p and $p+1$ differ, and the merge identifies them only if $p\equiv m-1$ and $p+1\equiv 1$, which forces $\delta \mid m-1$, or $p\equiv 1$ and $p+1\equiv m-1$, which forces $\delta \mid m-3$. (b) (`circulant_sweep`.) For the offset $m-k$ the role of δ is played by $\delta'=k-1$, and $m-1=\delta+\delta'+1$, $m-3=\delta+\delta'-1$. If the condition fails for both offsets, then $\delta \mid \delta'\pm 1$ and $\delta' \mid \delta\pm 1$ with $\delta,\delta'\ge 2$, which forces $|\delta-\delta'|=1$ and then $2k=m\pm 1$. □

With the mirror sweep, if $2k\not\equiv \pm 1 \pmod m$*, then* $N[K]$ *has an elimination certificate.* Indeed, by part (b) the condition of part (a) holds for the offset k, and part (a) applies, or for the offset $m-k$. In the second case relabel $k'_j=k_{-j}$: the relabelling $j\mapsto -j$ turns the arcs $k_{j\pm1}\to k_j$ into arcs of the same form and the chords $k_{j-k}\to k_j$ into chords $k'_{j-(m-k)}\to k'_j$ of offset $m-k$, so part (a) applies to the list $k'_0,k'_1,\dots,k'_{m-1}$, that is, to $k_0,k_{m-1},\dots,k_1$. This is the *mirror sweep*; it is not a single Lean declaration, and the formal proof of Corollary CC below realizes it by indexing cosets with powers of s^{-1}. The condition $2k\not\equiv 0$, which makes the chords one-way, is not needed. In a TT-free circulant $\mathrm{Cay}(\mathbb{Z}_m,\{1,-1,k\})$ with one-way chords the hypotheses hold, since chords of length 2 and chords with $2k\equiv\pm1$ close transitive triangles.

Theorem 5.10 (Theorem PC: blocks with private outside parents)**.** Lean: `ChordedPath`, `ChordedPath.elimOn`, `rateBound_pc` [`PathCert.lean`]
Lean: `ChordlessSubcubic`, `localGraph`, `chordless_block_decomp` [`Blocks.lean`]
Lean: `block_decomp_of_cycles` [`N3.lean`]

Let N be loopless and let V be partitioned into blocks. Suppose:

(P1) every vertex has at most one parent outside its block;

(P2) every vertex is a parent of at most one vertex of each other block;

(P3) every block K is

- *(a)* chordless subcubic*: every arc inside K belongs to a digon, every vertex of K has at most three parents, and $N[K]$ is TT-free; or*
- *(b) a chorded digon path with $N[K]$ TT-free; or*
- *(c) such that $N[K]$ has an elimination certificate.*

Then N satisfies ($\star$); consequently every storage code C for N over a nonempty finite alphabet satisfies $|C|^2\le|\alpha|^{|V|}$ (Theorem 2.2). Out-degrees are not restricted.

Outline of the formal proof. By Lemma S, blocks of kind (b) have elimination certificates and are therefore of kind (c) (`ChordedPath.elimOn`). The blocks are taken as pieces in an arbitrary order (`rateBound_pc`), each with the earlier blocks as context. By (P1) and (P2), every vertex of a block K has at most one parent outside K, and outside parents are private. A block of kind (c) is a piece by Lemma 5.6. A block of kind (a) is decomposed relative to its context by the shortest-cycle procedure of Theorem N3 (`chordless_block_decomp`, `block_decomp_of_cycles`), applied to its local graph (`localGraph`), which joins each vertex of K to its parents. Inside K the children of a vertex are its digon partners, so every vertex of K has local degree at most three; a vertex outside K has at most one local neighbour, by (P2); and the local graph has no triangle inside K, since such a triangle would consist of digons and contain a transitive triangle. The cycle pieces and the final relative 2-colouring then have private outside parents. □

The blocks are usually far from minimum pieces; Theorem PC is not a shortest-cycle argument. Its natural blocks are the components of the digon graph.

Corollary 5.11 (Theorem PC for digon components)**.** Not formalized; a complete proof is given below. *Let N be loopless and TT-free. Suppose that*

(H1) every vertex has at most one one-way parent, and

(H2) no vertex has one-way arcs to two vertices of the same component of the digon graph $G_2(N)$,

and that every component K of $G_2(N)$ has no one-way arc inside it and only vertices with at most three parents, or induces a path of $G_2(N)$, or is such that $N[K]$ has an elimination certificate (for instance a cycle of $G_2(N)$ of length m whose one-way arcs inside K all have the same offset k with $2k \not\equiv \pm 1 \pmod m$, by Proposition 5.9 with the mirror sweep; TT-freeness gives $3 \le k \le m-3$). Then N satisfies $(\star)$.

Proof (not formalized). Take the components of $G_2(N)$ as blocks. The digon partners of a vertex lie in its own component, so its parents outside its component are one-way; by (H1) there is at most one, which is (P1). A vertex outside a component K has only one-way arcs into K, at most one by (H2), which is (P2). For (P3): a component without one-way arcs inside it has every internal arc in a digon, and with the in-degree bound and TT-freeness it is of kind (a). If K induces a path $k_0, \dots, k_{L-1}$ of $G_2(N)$, its consecutive pairs are digons, and every other arc inside K is one-way, because a digon inside K is an edge of $G_2(N)[K]$, which is the path. Each vertex is the head of at most one one-way arc, by (H1), and the tail of at most one one-way arc into K, by (H2); so K is a chorded digon path, of kind (b). The remaining components are of kind (c) by hypothesis. Theorem PC applies. □

Every chordless cycle of $G_2(N)$ is of kind (a), and also of kind (c) by Lemma 5.7(b); it has length at least four, since $G_2(N)$ is triangle-free. Whether every TT-free chorded digon cycle is of kind (c) is Conjecture 9.2.

Why it matters. Theorem PC covers relations that none of Theorems N3, U3 and T covers, and puts no bound on out-degrees: a vertex may send one one-way arc into each of arbitrarily many blocks. For (T), side information built from two-way exchanges inside blocks, linked by one-way arcs that are private in the sense of (P1) and (P2), gives no coding gain. With the digon components as blocks, a construction needs, on a large scale, vertices that receive several one-way arcs or send several into one component, or components without certificates. The relation X_{15} of Remark 7.3 has vertices of the first kind, and yet it has a global certificate.

5.4 Cayley relations: Corollary CC

Corollary 5.12 (Corollary CC: Cayley relations)**.** `Lean`: `cayN`, `coset_steps`, `mem_coset_iff`, `rateBound_cayley`, `card_sq_le_cayley` [`Cayley.lean`]

Let G be a finite group and let $s, t \in G$ be such that $S = \{s, s^{-1}, t\}$ has three elements and does not contain 1. Let N be the Cayley relation with arcs $g \to gx$ for $x \in S$, so $N(q) = \{qs^{-1}, qs, qt^{-1}\}$. If N is TT-free, then N satisfies $(\star)$, and every storage code C for N over a nonempty finite alphabet satisfies $|C|^2 \le |\alpha|^{|G|}$.

Outline of the formal proof. The relation is loopless because $1 \notin S$. If $t^2 = 1$, every arc is reciprocated, the underlying graph has degree at most three, and Theorem U3 applies. Otherwise take the

left cosets $g\langle s\rangle$ as blocks. The order m of s is at least four: $m \neq 1, 2$ because $s \neq 1$ and $s \neq s^{-1}$, and $m \neq 3$ because $g \to gs \to gs^2 = gs^{-1}$ with $g \to gs^{-1}$ would be a transitive triangle. The only parent of q outside its coset is qt^{-1}, and the only possible child outside it is qt; so (P1) and (P2) hold. Index the coset of g by $j \mapsto gs^j$ (`mem_coset_iff`); in-neighbours through $s^{\pm 1}$ then sit at neighbouring positions (`coset_steps`). If $t \notin \langle s\rangle$, its internal arcs join consecutive positions, and Lemma 5.7(b) applies. If $t = s^k$, its internal arcs are $gs^{j\pm 1} \to gs^j$ and $gs^{j-k} \to gs^j$. Here $k \neq 0, \pm 1$ because $|S| = 3$ and $1 \notin S$; $k \neq \pm 2$ and $2k \not\equiv \pm 1$ because each would close a transitive triangle. So $3 \leq k \leq m - 3$ and Proposition 5.9 applies, with the generator s or, for the mirror sweep, s^{-1} and the offset $m - k$. Theorem PC concludes. □

If instead S has at most two elements and does not contain 1, every vertex has at most two parents, and Theorem T applies when N is TT-free. For example, $\mathrm{Cay}(\mathbb{Z}_9, \{1, -1, 3\})$, with $N(q) = \{q - 1, q + 1, q - 3\}$, is covered by Corollary CC (one block, $m = 9$, $k = 3$, $\delta = 5$). It is covered by none of the earlier results: its underlying graph is 4-regular, so neither Theorem N3, even with Lemma 2.4, nor Theorem U3 applies; it is strongly connected with every in-degree equal to three, so Theorem T and Corollary T_{int} do not apply; and it is not 2-dicolourable, since the two ends of a digon must lie in different acyclic sets and its digons form a 9-cycle. It also shows that the private pieces of Theorem N3 can fail to exist (Proposition 7.1).

Why it matters. Cayley relations are the side information of uniform, translation-invariant schedules. For every finite group, the pattern "two-way exchange along one generator, one-way transfer along another" gives no coding gain at any size. For (T), symmetric constructions of this shape are excluded, whatever the group.

6 Barrier bounds, guessing numbers and index codes

The relative form of Theorem D gives exact values on triangle-free graphs of maximum degree three: the guessing number is the fractional matching number, and the broadcast rate of index coding is the number of vertices minus it. We end the section with a general consequence for relations that become one of our classes after deleting a few vertices.

For a graph G and $S \subseteq V$ let $I(S) = \{v \notin S : N_G(v) \subseteq S\}$, the set of isolated vertices of $G - S$ (`isolatedOut`).

Theorem 6.1 (Barrier bound)**.** Lean: `isolatedOut`, `barrier_ineq`, `log_card_le_barrier` [`Barrier.lean`]

Let G be a triangle-free graph of maximum degree at most 3 and $S \subseteq V$. Every polymatroid h valid for N_G satisfies

$$2h(V) \;\leq\; 2\sum_{v\in S} h(v) + \sum_{v\notin S\cup I(S)} h(v),$$

and every storage code C for G over a nonempty finite alphabet α satisfies $2\log|C| \leq (|V| + |S| - |I(S)|)\log|\alpha|$.

Outline of the formal proof. Let $U = S \cup I(S)$. Theorem N3 gives a decomposition from the context U (`n3_decomp`), and the relative form of Theorem D gives $2(h(V) - h(U)) \leq \sum_{v\notin U}(h(U \cup \{v\}) - h(U)) \leq \sum_{v\notin U} h(v)$. The vertices of $I(S)$ have all their neighbours in S, so $h(U) = h(S) \leq \sum_{v\in S} h(v)$ by the closure lemma and subadditivity. For codes, apply this to the entropy of a uniformly random codeword, as in Theorem 2.2. □

Definition 6.2 (Fractional matchings)**.** Lean: `IsFracMatching`, `fracWeight`, `fracMatchingNumber` [`Matching.lean`]

A *fractional matching* of a graph G is a symmetric function $y\colon V \times V \to \mathbb{R}_{\geq 0}$ that vanishes on non-adjacent pairs and satisfies $\sum_u y(v,u) \leq 1$ for every v. Its *weight* is $\frac{1}{2}\sum_{u,v} y(u,v)$, each edge counted once. The *fractional matching number* $\nu^*(G)$ is the supremum of the weights.

Proposition 6.3 (The fractional Tutte–Berge formula)**.** Lean: `fracWeight_le_barrier`, `exists_doubleCover_matching`, `dcFrac`, `dcFrac_isFracMatching`, `fracWeight_dcFrac`, `fracMatchingNumber_spec`, `fracMatchingNumber_eq_min_barrier` [`Matching.lean`]

For every finite simple graph G, the supremum $\nu^*(G)$ is attained by a half-integral fractional matching, and

$$2\nu^*(G) \;=\; \min_{S \subseteq V} (|V| + |S| - |I(S)|).$$

The formula is classical (see [29]); we formalize it because the next theorem needs both directions.

Outline of the formal proof. Weak duality (`fracWeight_le_barrier`): the vertices of $I(S)$ send all their weight into S, which receives at most $|S|$, and every other vertex carries weight at most one, so twice the weight is at most $|S| + (|V| - |I(S)|)$. For the converse, take a set X_0 of largest deficiency $d = |X_0| - |N(X_0)|$, where $N(X)$ is the set of neighbours of X. Hall's theorem [23], in its Mathlib form, applied to the bipartite double cover of G with d dummy vertices added on one side, gives a set $D \subseteq V$ and an injective map g on D sending each vertex to a neighbour, with at most d vertices outside D (`exists_doubleCover_matching`). Putting weight $\frac{1}{2}$ on each pair $\{v, g(v)\}$, once per direction, gives a half-integral fractional matching of weight $|D|/2$ (`dcFrac`, `fracWeight_dcFrac`). For $S = N(X_0) \setminus X_0$ the set $I(S)$ contains $X_0 \setminus N(X_0)$, which gives $|D| \geq |V| + |S| - |I(S)|$. With weak duality, both bounds are equalities (`fracMatchingNumber_spec`). □

Theorem 6.4 (The guessing number is the fractional matching number)**.** Lean: `log_card_le_fracMatchingNumber`, `exists_code_fracMatchingNumber`, `guessingNumber_eq_fracMatching` [`Matching.lean`]

(a) For every finite simple graph G and every nonempty finite set β, some storage code for G over the alphabet $\beta \times \beta$ has at least $|\beta \times \beta|^{\nu^*(G)}$ words.

(b) If G is triangle-free of maximum degree at most 3, every storage code for G over every nonempty finite alphabet α has $|C| \leq |\alpha|^{\nu^*(G)}$, and over every alphabet $\beta \times \beta$ some storage code has exactly $|\beta \times \beta|^{\nu^*(G)}$ words.

The Lean statements are logarithmic, for instance $\nu^*(G)\log|\beta \times \beta| \leq \log|C|$, with the convention $\log 0 = 0$. They give the counts above whenever $\nu^*(G) > 0$ and $|\beta| \geq 2$; in the remaining cases the claims about $\beta \times \beta$ ask only for one word, and a code with a single word is a storage code.

Outline of the formal proof. (a) Take D and g as in Proposition 6.3. The code consists of the words x in which, for every $v \in D$, the second half of x_v equals the first half of $x_{g(v)}$, and all other halves are a fixed symbol: each vertex of D shares one half of its symbol with its neighbour $g(v)$, and each vertex of $g(D)$ receives one half from its preimage. Every symbol is then determined by the neighbours, and the $|\beta|^{|D|} = |\beta \times \beta|^{\nu^*}$ choices of the shared halves give distinct words (`exists_code_fracMatchingNumber`). (b) Theorem 6.1 with a set S attaining the minimum of Proposition 6.3 (`log_card_le_fracMatchingNumber`); equality by (a). □

Remark 6.5 (Guessing numbers; not formalized)**.** Let G be triangle-free of maximum degree at most three, and let $q \geq 2$. By Theorem 6.4(b), $\mathrm{gn}(G,q) \leq \nu^*(G)$. If $q = r^2$ is a perfect square, the code of part (b) over $\beta \times \beta$ with $|\beta| = r$ has $q^{\nu^*(G)}$ words, so $\mathrm{gn}(G,q) = \nu^*(G)$. Hence the asymptotic guessing number $\mathrm{gn}(G) = \sup_q \mathrm{gn}(G,q)$ equals $\nu^*(G)$. The restriction to squares cannot be dropped: by a theorem of Atkins, Rombach and Skerman [2], for every $q \geq 2$ that is not a perfect square and every sufficiently large odd n, the odd cycle C_n, which lies in this class and has $\nu^*(C_n) = n/2$, has $\mathrm{gn}(C_n,q) = (n-1)/2 + \log_q a < n/2$, where a is the largest divisor of q that is at most $\sqrt{q}$.

The code of part (a) realizes Christofides and Markström's fractional clique cover strategy [19], restated in storage language by Mazumdar, McGregor and Vorotnikova [27]: on a triangle-free graph the cliques are the vertices and the edges, and the fractional clique cover number is $|V| - \nu^*(G)$ (Remark 6.8). Christofides and Markström conjectured that this strategy is optimal on every undirected graph (see [20, Conjecture 1]) and proved it for perfect graphs and odd cycles [19]; Mazumdar, McGregor and Vorotnikova proved it for prisms and certain chorded cycles [27]. The conjecture is false in general: Baber, Christofides, Dang, Vaughan and Riis disproved it [3] (Dang's thesis gives a counterexample on ten vertices [20, Section 4.1]), and Cameron, Dang and Riis [18] found triangle-free counterexamples, such as the Higman–Sims graph. For the Clebsch graph, of degree five, they proved $10 \leq \mathrm{gn} \leq 11$, and the fourth paper determined $\mathrm{gn} = 10 > 8 = \nu^*$ [11, Corollary 6.11]. Theorem 6.4 proves the conjecture for triangle-free graphs of maximum degree three.

Definition 6.6 (Index codes)**.** `Lean: IndexCode` [`IndexCoding.lean`]
An *index code* for N over an alphabet α with a finite message set M is an encoding map $E\colon \alpha^V \to M$ together with decoders D_v, $v \in V$, such that $D_v(E(x), x) = x_v$ for every word x, where $D_v(m,x)$ depends only on m and on the restriction of x to $N(v)$.

Theorem 6.7 (Index codes and fractional matchings)**.** `Lean: IndexCode.isStorageCode_fibre, log_card_message_ge, exists_indexCode` [`IndexCoding.lean`]

(a) If G is triangle-free of maximum degree at most 3, every index code for N_G over a nonempty finite alphabet α has $\log|M| \geq (|V| - \nu^*(G)) \log|\alpha|$.

(b) For every finite simple graph G and every finite abelian group β, there is an index code for N_G over $\beta \times \beta$ with $\log|M| = (|V| - \nu^*(G)) \log|\beta \times \beta|$.

Outline of the formal proof. (a) The words with a common message form a storage code (`IndexCode.isStorageCode_fibre`), as in the fourth paper [11, Remark 4.9(b)]; each has at most $|\alpha|^{\nu^*}$ words by Theorem 6.4, and the $|M|$ fibres cover α^V. (b) With D and g as in Proposition 6.3, broadcast, for every v, the difference between the second half of x_v and the first half of $x_{g(v)}$ when $v \in D$, and the second half of x_v otherwise, and broadcast the first halves of the vertices outside $g(D)$. A vertex recovers the second half of its symbol from its message and the first half of $x_{g(v)}$, and, if $v = g(u)$, the first half from the message of u and the second half of x_u; both $g(v)$ and u are neighbours. The message set has $|\beta|^{2|V|-|D|}$ elements. □

The duality between storage codes and index codes behind part (a) is that of Mazumdar [26], Shanmugam and Dimakis [30], and Arbabjolfaei and Kim [1].

Remark 6.8 (The broadcast rate and fractional clique covers; not formalized)**.** Let G be triangle-free of maximum degree at most three. For $t \geq 1$ let β_t be the least number of bits, $\lceil \log_2 |M| \rceil$, of an index code for N_G over the alphabet $\{0,1\}^t$; the broadcast rate of [8] is $\lim_t \beta_t/t$. Part (a)

of Theorem 6.7, with $\alpha = \{0,1\}^t$, gives $\beta_t \geq (|V| - \nu^*(G))\, t$. Part (b), with $\beta = \mathbb{F}_2^t$, gives an index code over $\{0,1\}^{2t}$ with exactly $2(|V| - \nu^*(G))\, t$ bits, an integer because $2\nu^*(G)$ is one; so $\beta_{2t} \leq 2(|V| - \nu^*(G))\, t$. Hence the broadcast rate of G is $|V| - \nu^*(G)$.

This number is the fractional clique cover number of G, the least total weight of weighted cliques, here vertices and edges, that cover every vertex at least once. A fractional matching x gives the cover with weight x_e on each edge and $1 - \sum_{e \ni v} x_e$ on each vertex v, of total weight $|V| - \sum_e x_e$. Conversely, in a cover, lowering the weight of a vertex that is covered more than once, or lowering the weight of an edge uv at such a vertex v and raising that of the vertex u by the same amount, keeps a cover and does not increase its weight; so some optimal cover covers every vertex exactly once, and then its edge weights form a fractional matching whose weight is $|V|$ minus that of the cover. The bound of part (b) is thus the fractional clique cover upper bound on the broadcast rate [8], realized over $\beta \times \beta$ by a double-cover matching. For some graphs of the class these values were known: Blasiak, Kleinberg and Lubetzky determined the broadcast rate of cycles and of the cubic circulants, and computed that of the Petersen graph numerically [8].

Why it matters. For triangle-free subcubic side information, Theorems 6.1, 6.4 and 6.7 settle the storage and index-coding problems exactly, for all codes and all alphabets. In the language of the fourth paper's crux (Section 8), on such side information an index code must broadcast at least $|V| - \nu^*(G) \geq |V|/2$ symbols, and the extremal codes are fractional pairings: there is no room for a coding gain.

6.1 Relations near a class

All our classes are defined by exact conditions. A few exceptional vertices cost only what they could carry.

Corollary 6.9 (Paid deletion)**.** Not formalized; a complete proof is given below.

Let N be a recovery relation on V and $Z \subseteq V$. If $N - Z$ satisfies $(\star)$, then every storage code C for N over a nonempty finite alphabet α satisfies $|C|^2 \leq |\alpha|^{|V|+|Z|}$; that is, $\log_{|\alpha|}|C| \leq (|V| + |Z|)/2$.

Proof (not formalized). The paid-conditioning lemma [15, Lemma 2.2], Lean-verified there, says that every fibre of C obtained by fixing the values on Z and restricting to $V \setminus Z$ is a storage code for $N - Z$, and that if every storage code for $N - Z$ has at most $|\alpha|^u$ words, then $|C| \leq |\alpha|^{|Z|+u}$. (Directly: two words of a fibre that agree on $N(v) \setminus Z$, for $v \notin Z$, agree on $N(v)$, since they agree on Z, hence at v; and there are at most $|\alpha|^{|Z|}$ fibres.) By Theorem 2.2 we may take $u = (|V| - |Z|)/2$. □

Combined with Lemma 2.1, the corollary shows that relations with few short paired paths are as good as TT-free ones: if deleting the loops, the middle vertices of transitive triangles and a further set Z' leaves a relation in one of our classes, then storage codes have rate at most $\frac{1}{2} + (b_E(3) + |Z'|)/(2|V|)$.

Why it matters. This is how the exact theorems of this paper apply to the asymptotic setting of the paired-distance question [15, Question 9.1]: a construction must keep its cone relations away from every class here by a linear number of vertices (Section 8).

7 Limits of the method

The decompositions above are produced by rules: contract a shortest cycle (Theorems N3 and U3), a shortest directed cycle of in-degree-two vertices (Theorem T), or a whole block (Theorem PC).

This section shows where such rules stop. Only Proposition 7.1 is a theorem. The remarks report computations, labelled as numerical, and a cited formula; none of them is used in a proof. The computations are reproduced by the script `tools/verify_remarks.py`, described in `tools/README.md`.

7.1 Private pieces need not exist

The induction of Theorem N3 removes a piece whose outside parents are private. In directed relations of degree three such pieces need not exist, even when the half bound holds.

Proposition 7.1 (No private proper pieces)**.** Not formalized; a complete proof is given below.
Let N be the relation of $\mathrm{Cay}(\mathbb{Z}_9, \{1, -1, 3\})$*, so $N(q) = \{q-1, q+1, q-3\}$ and the children of q are $q+1$, $q-1$ and $q+3$. There is no set H with $\emptyset \neq H \neq \mathbb{Z}_9$ such that*

(a) every vertex of H has at most one parent outside H, and

(b) every vertex outside H is a parent of at most one vertex of H.

Proof (not formalized). Suppose H exists. Since $\emptyset \neq H \neq \mathbb{Z}_9$, some $a \in H$ has $a-1 \notin H$. By (a) at a, two of its parents $a-1, a+1, a-3$ lie in H, so $a+1 \in H$ and $a-3 \in H$. The children of $a-1 \notin H$ are $a-2$, a and $a+2$; since $a \in H$, (b) gives $a-2 \notin H$ and $a+2 \notin H$. Now the parents of $a+1$ are a, $a+2$ and $a-2$, and only a lies in H, contradicting (a). The residues $a-3, \ldots, a+2$ are distinct modulo 9, so the argument is valid. □

Every piece of a decomposition in context $\emptyset$ satisfies (a), and a piece whose outside parents are private satisfies (b). So on this relation a decomposition must start with the whole vertex set or with a piece whose outside parents are shared. Both happen. Corollary CC gives a single piece; explicitly, the records $\{0,4\}$, $\{1,3,6,8\}$, $\{2,5,7\}$ with the list $0,1,2,3,4,7,5,6,8$ form a global elimination certificate. And nested charges give a proper first piece, as follows.

Example 7.2 (A proper piece with shared outside parents)**.** In the relation of Proposition 7.1, let $H_1 = \{0,1,2,3,5,6\}$ with records $X = \{0,2,6\}$, $Y = \{1,3,5\}$ and an empty record Z, and the list $2,3,5,6,0,1$ with supplying records Y, X, X, Z, Y, Z. The outside parents are 8 for the vertices 2 and 0, 4 for 3 and 5, and 7 for 6 and 1; each outside vertex is shared by two steps with the same supplying record, so the certificate is co-supplied (Lemma 3.5). For instance, at the step of 5 the revealed set is $\{2,3\}$ and $N(5) = \{4,6,2\}$ lies in $\{2,3\} \cup X \cup \{4\}$. The relation that remains, on $\{4,7,8\}$ with H_1 as context, is decomposed by the pieces $\{4\}$ and $\{7,8\}$.

Why it matters. For an impossibility proof, Proposition 7.1 shows that the private-piece induction of Theorem N3 cannot prove half bounds on directed side information of degree three, even on a single circulant: an argument must use whole blocks, as in Theorem PC, or outside parents shared by several steps, which is what the nested charging of Lemma E provides.

7.2 Minimum nonempty cores are not enough

Call $H \subseteq V$ a *core* if every vertex of H has at most one parent outside H. A piece in context $\emptyset$ is a core, and in a cubic graph the minimum nonempty cores are the shortest cycles, which is why Theorem N3 can always contract a minimum nonempty core. In directed relations this fails.

Remark 7.3 (Numerical: a 15-vertex relation whose minimum nonempty core is not a piece)**.** Let X_{15} be the relation on $\{0, \ldots, 14\}$ with

q	0	1	2	3	4	5	6	7
$N(q)$	$\{5,1,6\}$	$\{0,2,7\}$	$\{1,3,8\}$	$\{2,4,6\}$	$\{3,5,7\}$	$\{4,0,8\}$	$\{14,0,10\}$	$\{13,4,9\}$
q	8	9	10	11	12	13	14	
$N(q)$	$\{9,14,5\}$	$\{6,11,7\}$	$\{9,12\}$	$\{2,13,12\}$	$\{14,11\}$	$\{1,8,11\}$	$\{3,13,12\}$	

A computation (`tools/verify_remarks.py`) shows: X_{15} is loopless and TT-free, every vertex has at most three parents and at most three children, it is strongly connected, and it is not 2-dicolourable. Its unique minimum nonempty core is $C = \{0,\dots,5\}$, a bidirected 6-cycle whose outside parents 6, 7, 8 each have two antipodal children: $6 \to 0,3$, $7 \to 1,4$ and $8 \to 2,5$. An exhaustive search over the partitions of C into at most seven records, all elimination orders and all supplying records finds no piece certificate for C in context $\emptyset$, even when the steps charged to an outside parent are only required to be nested (co-supplied outside parents are a special case). More records cannot help: C has six vertices, so a certificate with more than seven records has at least two empty records, which can be merged. So C is not a piece in context $\emptyset$, with any number of records. Nevertheless X_{15} has a decomposition, with a single piece: a global elimination certificate with three records, obtained by giving the vertices $0,\dots,14$ the records $0,1,0,1,0,1,1,1,0,0,0,1,0,0,2$ and using the list $0,4,5,7,1,9,8,2,11,12,10,6,3,13,14$, which can be checked by hand. So ($\star$) holds for X_{15}, but no rule that starts by contracting a minimum nonempty core can prove it.

In the undirected setting the antipodal pattern cannot occur on a shortest cycle: an outside vertex adjacent to c_0 and c_3 on a 6-cycle closes the 5-cycle w, c_0, c_1, c_2, c_3. In X_{15} the arc $6 \to 3$ is one-way ($0 \leftrightarrow 6$ is a digon); the 5-cycle $6,0,1,2,3$ of the underlying graph is not a core, since vertex 6 has the two parents 10 and 14 outside it, and the exhaustive search finds no nonempty core with fewer than six vertices. The relation lies outside Theorems U3 and T: vertex 6 has five neighbours in the underlying graph, and the vertices with three parents include the digons of C. Its digon components violate (H1) of Corollary 5.11, since vertex 6 has the two one-way parents 10 and 14; Theorem PC applies with the whole vertex set as one block, through the global certificate.

7.3 Sweeps and global certificates

For a TT-free chorded digon cycle $K = \{k_0,\dots,k_{L-1}\}$ (indices modulo L), the *sweep* from k_z in direction $d = \pm 1$ is the elimination list $k_z, k_{z+d}, \dots, k_{z+(L-1)d}$ with the finest records that make the later parents of each vertex lie in one record: the classes of the equivalence relation that merges these later parents, for every vertex. The sweep is *valid* if no vertex lies in the class of its later parents; it is then an elimination certificate. Proposition 5.9, with the mirror sweep, shows that a TT-free chorded digon cycle whose chords all have one offset k with $2k \not\equiv \pm 1 \pmod L$ has a valid sweep; this includes every TT-free circulant $\mathrm{Cay}(\mathbb{Z}_L, \{1,-1,k\})$.

Remark 7.4 (Numerical: chorded cycles without a valid sweep)**.** An adversarial local search over chord sets found twelve TT-free chorded digon cycles, of lengths 13, 15 and 16, none of whose $2L$ sweeps is valid (`tools/verify_remarks.py`). One of them has $L = 13$ and the chords

$$0 \to 8,\ 1 \to 6,\ 3 \to 11,\ 4 \to 0,\ 5 \to 10,\ 6 \to 9,\ 7 \to 3,\ 8 \to 4,\ 9 \to 12,\ 10 \to 1,\ 11 \to 7,\ 12 \to 2.$$

All twelve have elimination certificates with three records. For the cycle displayed, give the vertices $0,\dots,12$ the records $0,2,0,1,0,2,1,2,1,2,1,2,1$; the list $11,9,7,6,5,4,10,1,2,0,8,3,12$ is then an elimination certificate, which can be checked by hand. So these cycles are of kind (c) in Theorem PC; they only show that sweeps, the method of Proposition 5.9, do not cover all chorded cycles.

Remark 7.5 (Numerical: the Petersen graph has no global certificate)**.** Let N be the neighbourhood relation of the Petersen graph. In a global elimination certificate of N (Definition 3.8), written as an elimination list, the later neighbours of each vertex lie in one record, different from the record of the vertex; when the number of records is not bounded, an elimination order admits a certificate exactly when these forced equalities and inequalities of records are consistent, since the classes of the forced equalities can then serve as the records. An exhaustive search over all elimination orders of N, which keeps the forced equalities in a union–find structure and abandons an order as soon as a forced inequality fails, finds no consistent order (`tools/verify_remarks.py`). So N has no global elimination certificate, with any number of records. Theorem N3, by contrast, decomposes the Petersen graph into two pieces: its proof first contracts a 5-cycle, whose outside parents (the spokes) are private, and then the remaining 5-cycle. So decompositions are strictly stronger than global certificates, and Theorem D strictly stronger than Proposition 3.9.

7.4 Odd girth and degree

Remark 7.6 (Cited: folded cubes)**.** The folded $(2k+1)$-cube $FQ_{2k+1} = \mathrm{Cay}(\mathbb{F}_2^{2k}, \{e_1, \dots, e_{2k}, \mathbf{1}\})$, where $\mathbf{1}$ is the all-ones vector, has $N = 2^{2k}$ vertices, degree $2k+1$ and odd girth $2k+1$; for $k = 2$ it is the Clebsch graph. Over $\mathbb{F}_2$, the matrix $I + A$ has rank $\frac{1}{2}(2^{2k} - 2^k)$: this is the case of even dimension $2k$ of a proposition of Barg, Schwartz and Yohananov [5, Proposition 13], whose Cayley graph also has the zero vector as a generator and therefore has adjacency matrix $I + A$. Its kernel is a binary linear storage code of dimension $N/2 + 2^{k-1}$; for $k = 2, 3$ this is also checked numerically (`tools/verify_remarks.py`). So ($\star$) fails on the folded cubes, whose odd girth equals their degree, for every $k \geq 2$; it holds on other graphs whose odd girth equals their degree, such as the 5-regular Andrásfai graph $\mathrm{And}(5) = K_{14/5}$, of odd girth five (Theorem 4.2). In the symmetric case $\mathrm{dist}_E(L_p, R_p)$ is the length of the shortest odd closed walk through p, so here $b_E(r) = 0$ for $r < 2k+1$. Hence, for every odd $r \geq 3$, an exact half bound that assumes only $b_E(r) = 0$ can hold at most for maximum degree $r+1$: the folded $(r+2)$-cube violates it. The excess $2^{k-1} = \sqrt{N}/2$ is $o(N)$, which is consistent with the asymptotic form of the paired-distance question [15, Question 9.1].

Remark 7.7 (Heuristic: degree four)**.** In the undirected setting, the method needs each step to have at most one outside parent. If a vertex v of a shortest cycle has two outside neighbours p and p', the step that eliminates it costs $I(v; \{p, p'\} \mid Q)$, which is not a sum of charges on p and p' that telescope separately. This is a limitation of Lemma E, not evidence against the half bound at degree four; Theorem 4.2 covers graphs of circular chromatic number below three at every degree. The fourth paper shows that the local linear-algebraic necessary conditions on a smallest counterexample, checked inside tree-like regions, cannot give rate $\frac{1}{2} + o(1)$ at degree four or more [11, Proposition 6.28].

8 What this means for transposition

Our aim is to decide (T), and we would prefer a construction. This section says what the results above require of a linear-time transposer, and of an oblivious one in particular. Apart from Proposition 8.1, a conditional statement with a complete proof, it contains no new theorems: every claim is one of the results above, a result of an earlier paper of the series, or explicitly labelled as a heuristic.

8.1 The interface

For oblivious machines, the fourth paper's Theorem D and its Corollary 4.11 [11, Theorem 4.10 and Corollary 4.11] reduce ¬(T) to two statements: *depth reduction*, that the time–space dependency

graph of every oblivious linear-time run can be cut to bounded depth by a genie revealing εL bits; and *the crux*, that index codes for the side information of the remaining bounded-depth cones need more than εL bits [11, Definition 4.8]. Every fibre of an index code is a storage code for the cones' recovery relation N_I [11, Remark 4.9(b)]. Hence a correct oblivious transposer whose genie reveals εL bits yields a binary storage code of rate at least $1-\varepsilon$ on N_I; conversely, rate bounds $\frac{1}{2}+o(1)$ for all codes on such relations, together with depth reduction at some cost $\varepsilon < \frac{1}{2}$, would rule out oblivious linear-time transposers. Depth reduction is proved for home-based work heads [11] and, in the companion paper published with this one [10], for every family of head walks of bounded nesting load, which covers every machine whose work heads enter each cell boundedly often; it is open in general.

Directed Recovery Structures [15] describes the relations N_I: for fixed-depth cones of linear-time schedules, after deleting a small fraction of coordinates of high reuse (its Corollary 3.8), only $O(L/\log d)$ coordinates p have $\mathrm{dist}_E(L_p, R_p) \le r$, for every fixed r (its Theorem 3.3 and Corollary 3.7). Deleting these coordinates leaves a relation with $b_E(r) = 0$, since deleting coordinates does not shorten paired paths; for $r = 3$ it is loopless and TT-free (Lemma 2.1). Its Question 9.1 asks whether all codes on bounded-degree relations with $b_E(r) = o(|V|)$ for every fixed r have rate at most $\frac{1}{2}+o(1)$; by its Proposition 9.2, an affirmative answer gives this bound on all actual cone relations.

Our theorems are exact half bounds, at radius three, on classes of TT-free relations, several of them with no degree bound. With Corollary 6.9 they answer Question 9.1 affirmatively for every sequence of relations that, after deleting $o(|V|)$ coordinates, lies in one of our classes. They do not decide (T): we do not know that cone relations lie in these classes, and depth reduction is open. The degree-three case of Question 9.1 would follow from a single combinatorial statement.

Proposition 8.1 (What the directed degree-three conjecture would give)**.** Not formalized; a complete proof is given below.

Suppose that every loopless TT-free relation in which every vertex has at most three parents satisfies ($\star$)*; Conjecture 9.1 below would imply this, by Theorem D. Then every storage code on a relation N in which every vertex has at most three parents has rate at most $\frac{1}{2} + b_E(3)/(2|V|)$. In particular, [15, Question 9.1] would have an affirmative answer for relations with recovery sets of size at most three, with no bound on out-degrees and with the paired-distance condition used only at radius three.*

Proof (not formalized). Let Z be the set of coordinates p with $\mathrm{dist}_E(L_p, R_p) \le 3$, so $|Z| = b_E(3)$. It contains every vertex with a loop and the middle vertex of every transitive triangle (Lemma 2.1; the path L_p, R_z, L_x, R_p given by a transitive triangle $x \to p \to z$, $x \to z$ needs no looplessness). A transitive triangle of $N - Z$ is one of N, whose middle vertex lies in Z; so $N - Z$ is loopless and TT-free, and its vertices still have at most three parents. By hypothesis it satisfies ($\star$), and Corollary 6.9 gives $\log_{|\alpha|}|C| \le (|V| + b_E(3))/2$. □

8.2 Necessary conditions for a construction

Consider an oblivious linear-time transposer and a genie that reveals εL bits and leaves cones of bounded depth, with recovery relation N_I on the L input coordinates (for home-based work heads such genies exist for every $\varepsilon > 0$ [11, Theorem 4.20 and Corollary 4.23]). The machine yields a binary storage code of rate at least $1-\varepsilon$ on N_I. In addition to the requirements of the earlier papers of the series [12, 16, 11, 15], it must satisfy the following.

(N1) *No large induced sub-relation in our classes.* If W is a set of coordinates such that $N_I[W]$ satisfies ($\star$), then $|W| \le 2\varepsilon L$. Indeed, Corollary 6.9 with $Z = V \setminus W$ bounds the code by

$2^{L-|W|/2}$, while it has at least $2^{(1-\varepsilon)L}$ words. This applies to every W on which $N_I[W]$ is 2-dicolourable, or lies in one of the classes of Theorems U3, T, PC, Corollaries T_{int} and CC, or, when symmetric, of Theorems N3 and 4.2. For 2-dicolourable W this is a form of the classical bound of index coding by acyclic induced sub-relations [4]; the other classes are not covered by the acyclic bound.

(N2) *Where a gain must live.* For every fixed $\delta > 0$ there is a set Z of at most $\delta L + O(L/\log d)$ coordinates such that $N_I - Z$ is loopless and TT-free, with bounded degrees: the coordinates of high reuse and those with paired distance at most three [15, Theorem 3.3 and Corollaries 3.7 and 3.8]. By (N1), every induced sub-relation of $N_I - Z$ on more than $2\varepsilon L$ coordinates violates the hypotheses of each theorem of Section 5. So it must contain vertices with at least three parents inside their strongly connected component that lie on directed cycles among themselves (Theorem T and Corollary T_{int}); it must have underlying degree at least four somewhere (Theorem U3); and it admits no partition into blocks as in Theorem PC, in particular no elimination certificate of the whole sub-relation (with the digon components as blocks, this requires a vertex with two one-way parents, a vertex with two one-way arcs into one component, or a component of none of the three kinds). The relation X_{15} of Remark 7.3 has the first two features but not the third: it has a global certificate.

(N3) *Directed triangles do not help.* Native cone relations can contain linearly many disjoint directed 3-cycles [15, Proposition 3.10], which defeat symmetric triangle-free hypotheses. Our directed theorems allow them, so they are not by themselves a source of coding gain.

(N4) *Gains at short odd distance do not transfer.* The Clebsch graph and the folded cubes carry codes above rate $\frac{1}{2}$, and their odd girth does not exceed their degree (Remark 7.6; [11, Corollary 6.11]). After paid degree trimming, cone relations have only $O(L/\log d)$ coordinates with paired distance at most r, for every fixed r [15, Theorem 3.3], so such local structure can occupy only a vanishing fraction of a cone relation (heuristic reading: a gain must come from structure at growing paired distance, which is the open territory of Question 9.1).

8.3 Mechanisms that fail

The following classes of side information admit no storage code of rate above $\frac{1}{2}$, for any alphabet and any recovery functions; a construction cannot draw its coding gain from them, on more than a 2ε fraction of the coordinates.

- *Cycles through vertices with two internal inputs.* Relations whose strongly connected components have no transitive triangle and whose vertices with three or more internal parents induce an acyclic sub-relation, with any number of parents in earlier components and any out-degree (Corollary T_{int}). This contains the TT-free case of the two-input theorem of *Directed Recovery Structures* [15, Theorem 5.1].

- *Exchange blocks with private links.* Blocks of two-way exchanges, linked by one-way arcs that are private in the sense of (P1) and (P2), with blocks that are chordless subcubic, chorded paths or certified (Theorem PC); in particular every translation-invariant pattern $\mathrm{Cay}(G, \{s, s^{-1}, t\})$ without transitive triangles, on every finite group, when $\{s, s^{-1}, t\}$ has three elements and does not contain 1 (Corollary CC).

- *Subcubic directed structure.* Relations without transitive triangles whose underlying graph has maximum degree three, directed triangles included (Theorem U3).

- *Sparse or low-chromatic symmetric structure.* Symmetric side information of circular chromatic number below three, at any degree (Theorem 4.2), and triangle-free subcubic symmetric side information (Theorem N3), where the optimal codes are fractional pairings and index codes need $|V| - \nu^*$ symbols (Theorems 6.4 and 6.7).

For impossibility proofs, Section 7 records what fails as a method: private pieces (Proposition 7.1), minimum nonempty cores (Remark 7.3), sweeps (Remark 7.4) and global certificates alone (Remark 7.5). Whether nested decompositions always exist at in-degree three is Conjecture 9.1.

9 Open problems

9.1 Where the coding side stands

For oblivious machines, the series reduces ¬(T) to depth reduction and an all-code storage bound on the cone relations (Section 8.1). The table records the all-code half bounds now available at the exact radius three, that is, on loopless TT-free relations.

class	**result**	**open nearby**
symmetric, $\chi_c < 3$, any degree	circular memo, Theorem 4.2	symmetric, $\chi_c = 3$, degree ≥ 4
symmetric, triangle-free, degree ≤ 3	Theorem N3; gn $= \nu^*$ (Theorem 6.4)	symmetric, triangle-free, degree 4
underlying degree ≤ 3	Theorem U3	the all-code form of Theorem S of [11]
vertices with ≥ 3 internal parents acyclic	Theorem T, Corollary T_{int}	in-degree ≤ 3 in general (Conjecture 9.1)
blocks with private links	Theorem PC, Corollary CC	chorded digon cycles (Conjecture 9.2)

By Remark 7.6, an exact half bound that assumes only $b_E(r) = 0$ can hold at most up to maximum degree $r + 1$; at radius three, TT-freeness alone can give exact bounds at most up to degree four, and further classes need further structure, as in Theorems T and PC. The transposition question needs the asymptotic form, [15, Question 9.1].

9.2 Open problems

1. *Decide (T).* Construct a machine transposing $d \times d$ matrices with $b(d)$-bit entries in time $O(L)$, or prove that none exists. The first paper's frozen checker accepts either answer [12].

2. *The directed degree-three case.* The natural next class is all of in-degree three.

   Conjecture 9.1 (Conjecture P)**.** *Every loopless TT-free relation in which every vertex has at most three parents has a decomposition (Definition 3.3).*

   By Theorem D, Conjecture 9.1 implies that all such relations satisfy (⋆), and by Proposition 8.1 that the paired-distance question has an affirmative answer for recovery sets of size at most three, at any out-degree. Every relation treated in this paper has a decomposition, including X_{15} (Remark 7.3), and Remark 7.3 shows that a proof cannot always start from a minimum nonempty core. One intermediate target is the all-code version of the fourth paper's Theorem S on its whole class: TT-free relations with at most three parents per vertex, in which every

vertex with three parents has no child that is not a parent. It lies between Theorem U3 and Conjecture 9.1.

3. *Chorded digon cycles.*

   Conjecture 9.2 (Conjecture K)**.** *Let K be a chorded digon cycle, with at most one chord into and at most one chord out of every vertex, such that $N[K]$ is TT-free. Then $N[K]$ has an elimination certificate.*

   With Theorem PC, Conjecture 9.2 would give ($\star$) for every TT-free relation whose digon components are paths or cycles and which satisfies (H1) and (H2) of Corollary 5.11; this includes every TT-free relation in which every vertex has exactly two digon partners, one one-way parent and one one-way child. Proposition 5.9 with the mirror sweep proves it when all chords have one offset k with $2k \not\equiv \pm 1 \pmod{|K|}$, in particular for TT-free circulants; sweeps do not prove it in general (Remark 7.4), but every cycle of Remark 7.4 has a certificate with three records.

4. *Degree four.* Does every triangle-free graph of maximum degree four satisfy ($\star$), or at least have guessing number $|V|/2 + o(|V|)$ along families with growing girth? At degree five the exact bound fails (the Clebsch graph), and Remark 7.7 explains why Lemma E alone does not reach degree four. The circular memo's theorem gives the answer for $\chi_c < 3$ at every degree.

5. *The all-code paired-distance question* [15, Question 9.1]. Do bounded-degree recovery relations with $b_E(r) = o(|V|)$ for every fixed r admit only storage codes of rate at most $\frac{1}{2} + o(1)$, over every alphabet? With depth reduction for a class of oblivious schedules, an affirmative answer would exclude oblivious linear-time transposers in that class (Section 8.1); a counterexample would not by itself construct a transposer [15]. Our results answer it for relations that are close, in the sense of Corollary 6.9, to one of the classes above.

6. *Formalization.* Formalize the topological order of strongly connected components used in Corollary T_{int}, the digon-component form of Theorem PC (Corollary 5.11), Lemma 2.1, Corollary 6.9, Proposition 8.1 and Proposition 7.1, and turn the computations of Section 7 into Lean certificates.

A Index of Lean declarations

Every declaration of this paper's library cited in the text, grouped by the Lean module that contains it, relative to the namespace `HalfRate`. All of them depend only on the axioms `propext`, `Classical.choice` and `Quot.sound` (Appendix B). The declarations of the earlier libraries of the series that are named in the text, in the namespaces `Transpose` of the first paper, `MotionDepthCodes` of the fourth paper and `CircularMemo` of its circular memo, are defined and verified there [12, 11, 13].

`HalfRate.Barrier` `barrier_ineq`, `isolatedOut`, `log_card_le_barrier`.

`HalfRate.Blocks` `ChordlessSubcubic`, `ElimOn`, `ElimOn.piece`, `chordless_block_decomp`, `localGraph`.

`HalfRate.Cayley` `card_sq_le_cayley`, `cayN`, `coset_steps`, `mem_coset_iff`, `rateBound_cayley`.

HalfRate.Circulant circB, circCol, circulant_sweep, elimOn_circulant, elimOn_cycle, elimOn_ofFn.

HalfRate.Corollaries ElimCert, ElimCert.pullback, ElimCert.rateBound, IsAcyclicIn, andrasfaiCert, rateBound_of_circularHom', rateBound_of_twoAcyclic.

HalfRate.Decomposition Decomp, PieceCert, PieceCert.ineq, RelCharged, RelCharged.charged, card_sq_le_of_decomp, contr, contr_insert_eq, contr_isPolymatroid, decomp_ineq, rateBound_of_decomp.

HalfRate.IndexCoding IndexCode, IndexCode.isStorageCode_fibre, exists_indexCode, log_card_message_ge.

HalfRate.Matching IsFracMatching, dcFrac, dcFrac_isFracMatching, exists_code_fracMatchingNumber, exists_doubleCover_matching, fracMatchingNumber, fracMatchingNumber_eq_min_barrier, fracMatchingNumber_spec, fracWeight, fracWeight_dcFrac, fracWeight_le_barrier, guessingNumber_eq_fracMatching, log_card_le_fracMatchingNumber.

HalfRate.N3 CycIn, CycIn.chordless, CycIn.share, block_decomp_of_cycles, card_sq_le_n3, cyclePiece, decomp_of_cycles, n3_decomp, rateBound_n3.

HalfRate.NestedElimination Charged, defect, defect_eq_mi, elim_defect, mi_mono, nested_elimination, record_increment, sum_records_step, totalDefect, totalDefect_le.

HalfRate.PathCert ChordedPath, ChordedPath.elimOn, DownChord, elimOn_path, pathRoot, pathRoot_succ_ne, rateBound_pc.

HalfRate.Pieces PieceCert.mono, RelCharged.mono, bdirPiece, exists_relCharged, pieceCert_of_elim.

HalfRate.Positional relCharged_of_cosupply, rev.

HalfRate.TheoremT BlockOrder, DCyc, DCyc.acyclic_of_missing, DCyc.chordless, TTFreeOn, block_decomp, blocks_decomp, card_sq_le_thmT, exists_hasDCyc_of_sourceless, i5Piece, rateBound_blocks, rateBound_thmT, singlePiece.

HalfRate.U3 card_sq_le_u3, dtriPiece, dtri_orient, rateBound_u3, sub_nbhd_und, triPiece, und.

Also named in the text TTFree, mi.

B Building and verifying the formalization

The Lean project is the directory `lean/` next to this paper. It uses Lean and Mathlib, both at version v4.34.1 (`lean-toolchain`, `lakefile.toml`, `lake-manifest.json`). It requires, through relative paths, the Lean library of the fourth paper [11], in the directory `Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes/lean`, and that of its circular memo [13], in the subdirectory `memo/lean` of the same paper. The fourth paper's library in turn imports the second paper's library, which imports that of the first [12, 16], and the repository's shared library `lean-machine`. Keep the repository's directory layout, so that these sibling directories stay in place.

Build. With `elan` installed, run in `lean/`:

```
lake exe cache get && lake build
```

The first command downloads the compiled Mathlib; the second compiles the library and the parts of the earlier libraries that it uses. Before the first build, fetch the second paper's pinned external dependency as its README describes.

Axioms. The file `PaperAxioms.lean`, generated from the LaTeX sources of this paper by `tools/lean_index.py`, runs `#print axioms` on every declaration of this library named here (Appendix A):

```
lake env lean PaperAxioms.lean
```

Every declaration depends at most on `propext`, `Classical.choice` and `Quot.sound`. No Lean file of this library uses `sorry`, `admit` or `native_decide`, or declares an `axiom`. Running `python3 tools/lean_index.py -index-only` in the paper's directory regenerates only the index of Appendix A.

Where things are. `NestedElimination.lean` (Lemma E), `Decomposition.lean` (contraction, piece certificates and Theorem D), `Positional.lean` (co-supply), `Pieces.lean` (fewer arcs, greedy certificates, Lemma B_{dir}), `Corollaries.lean` (global certificates, the pull-back, 2-dicolourable relations and circular chromatic number below three), `N3.lean` (Theorem N3 and the shortest-cycle decomposition of blocks), `U3.lean` (Theorem U3), `TheoremT.lean` (Theorem T and the block form of Corollary T_{int}), `Blocks.lean` (internal certificates and chordless subcubic blocks), `PathCert.lean` (Lemma S and Theorem PC), `Circulant.lean` (positional certificates and Proposition CC), `Cayley.lean` (Corollary CC), `Barrier.lean` (the barrier bound), `Matching.lean` (fractional matchings, the fractional Tutte–Berge formula and the guessing number), `IndexCoding.lean` (index codes and the broadcast rate), and `Examples.lean` (small sanity checks).

Computations. The numerical remarks of Section 7 (Remarks 7.3 to 7.5) and the numerical check of the folded-cube formula in Remark 7.6 are reproduced by `tools/verify_remarks.py` (plain Python 3, run from the paper's directory as `python3 tools/verify_remarks.py`), described in `tools/README.md`; the script also checks Proposition 7.1 by exhaustion, and Example 7.2 and the global certificate before it step by step. They are not used in any proof, and the Lean library does not depend on them.

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